WELL WORKOVER PROGRAM

AZADEGAN OIL FIELD

WELL WORKOVER PROGRAM

AZN-10 (NEAR PAD #50), TYPE: IIF

Drilling Department

OCTOBR 2023

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun province on a large long-axis anticline with N-S trending with approximate dimensions about 20 km width and 60 km length near to Iran-Iraq border, to the east of Yaran Oil Field.

The trap area of the south Azadegan is limited to 43.2x17.2 km. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643m. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340m (Neyriz Fm.), mainly to evaluate the Jurassic formations.

Based on the exploration wells results, carbonate reservoir of Sarvak Formation is broadly developed in Azadegan Field. This formation was penetrated in all of seven exploration wells (AZN-1, AZN-2, AZN-3, AZN-4, AZN-5, AZN-6 and AZN-10). Clastic oil-bearing zones of Kazhdumi Formation developed in main anticline of Azadegan Field, extending from north to center of South Azadegan. The Clastic oil-bearing zones of Gadvan Formation developed only in crest area and south parts of South Azadegan Filed, which was discovered by AZN-1 in crest and AZN-4 in south parts. The carbonates of Fahliyan Formation also developed in Azadegan field, but the oil reservoir of this formation developed in South part of South Azadegan.

The Azadegan Field is an elongated anticline, trending in N-S direction, belonging to "Arabian Trend".

The stratigraphy sequence of the field includes AghaJari, Gachsaran, Fahliyan, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan Formations, from top to bottom. This sequence in well no AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus and Neyriz formations.

The Dominant lithofacies of Sarvak Formation are Rudist-bearing packstone and partially grainstone, which contains abundant bioclasts. Some other facies like matrix-dominated wackestone and mudstone are also found.

The AZADEGAN Oilfield is Located in Khuzestan Province and is Approximately Located 80 Kilometers West of Ahwaz City, near to Iran-Iraq Border. It is in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45ºc in summer, and Highest Temperature is 52ºc. The Lowest Temperature is -4.4ºc in winter. Maximum Rainfall is 78 mm in February, Winter Season.

AZN-10 well is located in the north part of Area II of South Azadegan Oil Field. The well is located near pad-50 nearby of AZNS-36 (S3). This well is abandoned and will be worked-over by de-suspension & 7" liner perforation in Fahliyan layer. Finally, the well will be completed in Fahliyan.

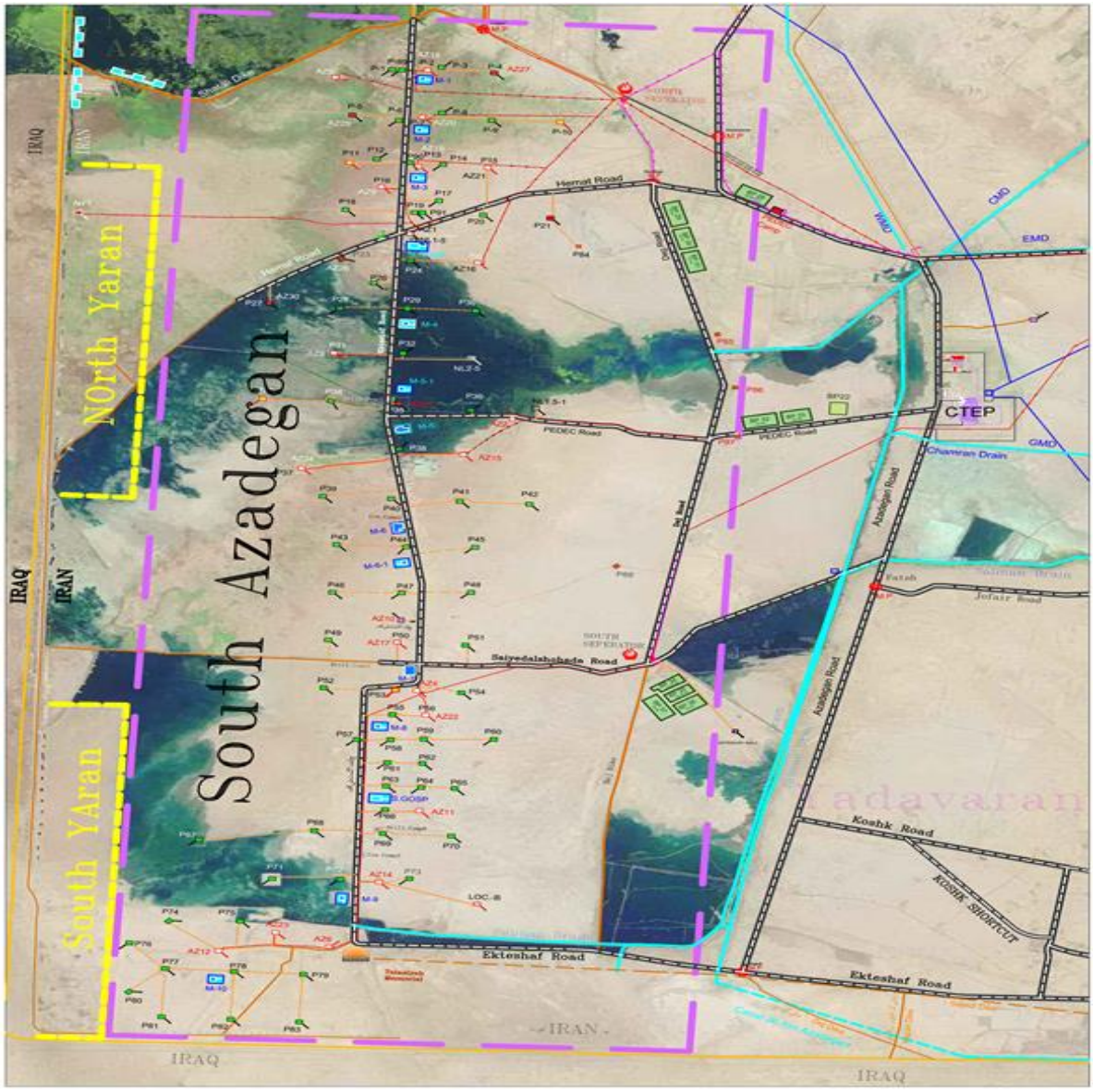


Figure 1: AZNS Oilfield Location Map

1.2 Well Information

Country / Area Name:	Iran / Khuzestan		
Client	Petro Pars Limited		
Field Name:	South Azadegan Oil Field		
Type Of Operation:	Workover		
Well Name / Type :	AZN-10 / Fahliyan well in Block II		
Pad Number:	Near 50		
Well Type:	Vertical		
Grid Geodetic system	UTM39N, WGS-84		
Wellhead Coordinates:	E: 194,371.6	N: 3,451,636.1	
E.P. in Fahliyan:	E: 194,377.2	N: 3,451,631.7	-3974.2m TVDss
TD Coordinates:	E: 194,387.9	N: 3,451,650.6	
Well Objective:	De-suspension & Completion/ Fahliyan Producer		
RTE:	13.45m		
GLE:	3.95m		
Well Duration :	39 (Main Plan) + 9 (Contingency Plan)		
Total Depth (MD):	5340 m		
Target Formation:	Fahliyan		
Completion Type:	7" LNR Cased-Cemented-Perforated		

Table 1: AZN-10 information

1.3 Area Distribution

According to the drilling history, the CONTRACT AREA is distributed into three blocks as per the geological features with regard to DRILLING OPERATION. This distribution may differ according to knowledge accumulated in the future WORK.
Contract area is as per below:

X (m)	Y (m)
185640	3434275
188020	3477361
205224	3476408
202842	3433327

Table 2: Area Distribution

Block I is located on the north part of the CONTRACT AREA in north of well AZN-10. Gachsaran Formation is characterized by abnormal high pressure in this block.
Block II is located between well AZN-10 and AZN-011. In this block, Gachsaran Formation is characterized by pressure tending to normal
Block III is located on the south of line well AZN-011. In this block, Gachsaran Formation is characterized by normal pressure and all drilled wells in this block suffered from complete loss at Pabdeh formation.

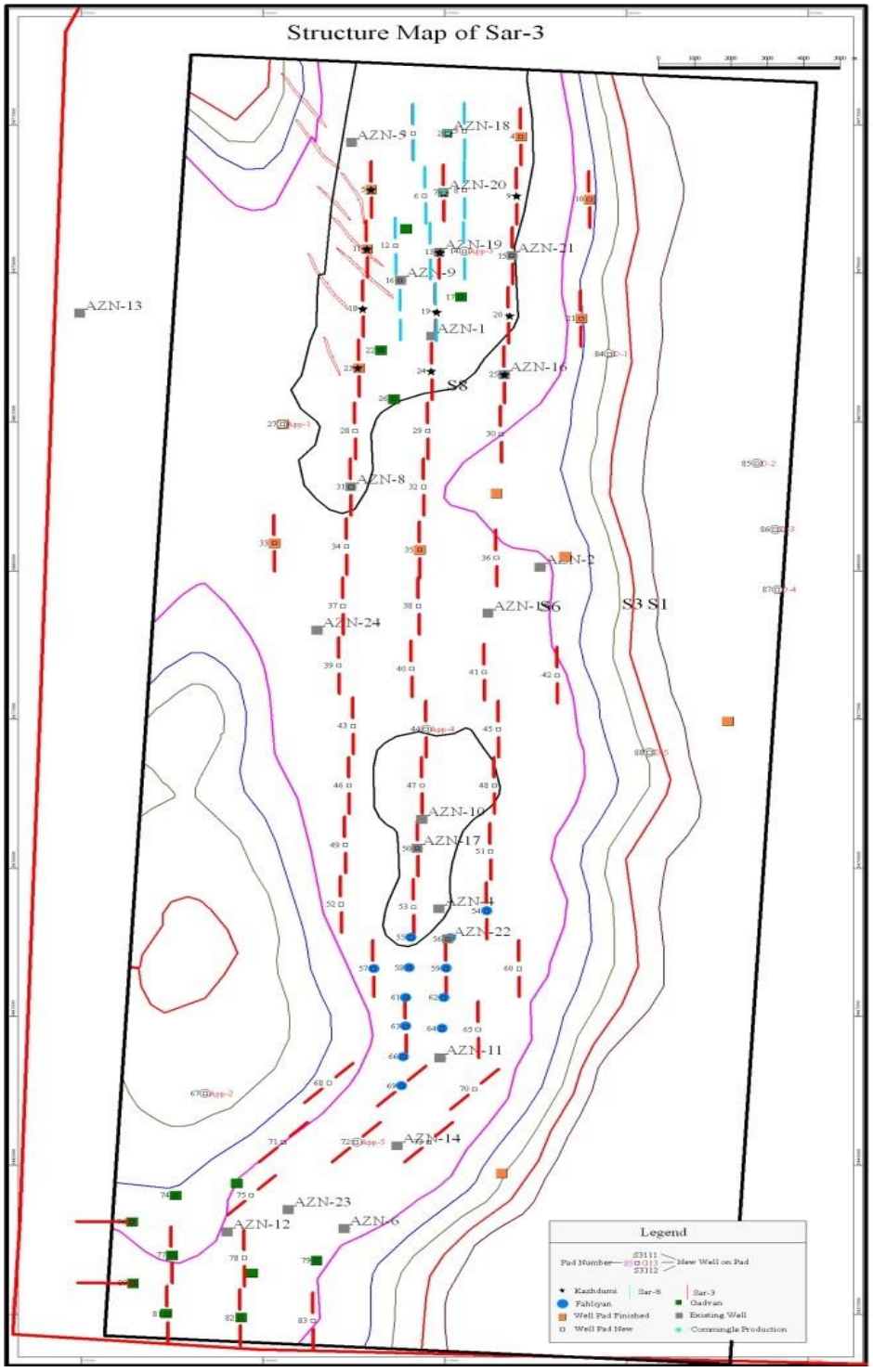


Figure 2: AZN Area Distribution

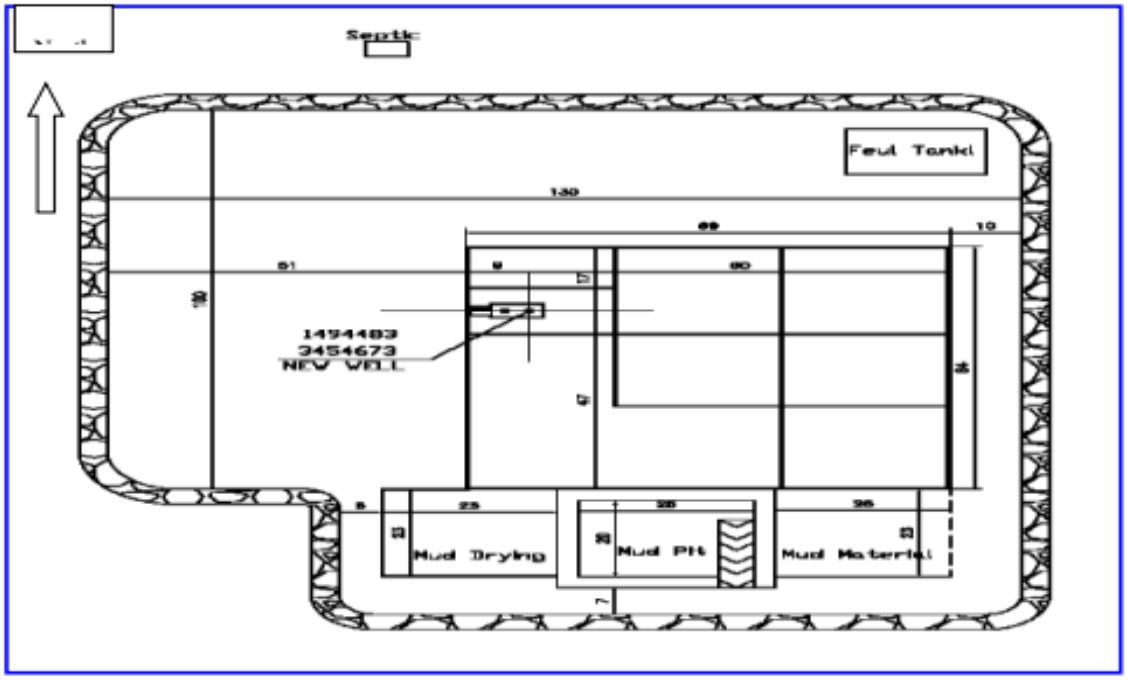
1.4 Well Pad Specification

Well Pad Type	Well Pad Dimension (m)	Mud Pit Dimension (m)	Cellar Dimension (m)	Conductor Size
Single Well	130 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D	30"
Double Well	145 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D	30"
Triple Well	160 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D + 4.0L x 3.8W x 3.4D	30'

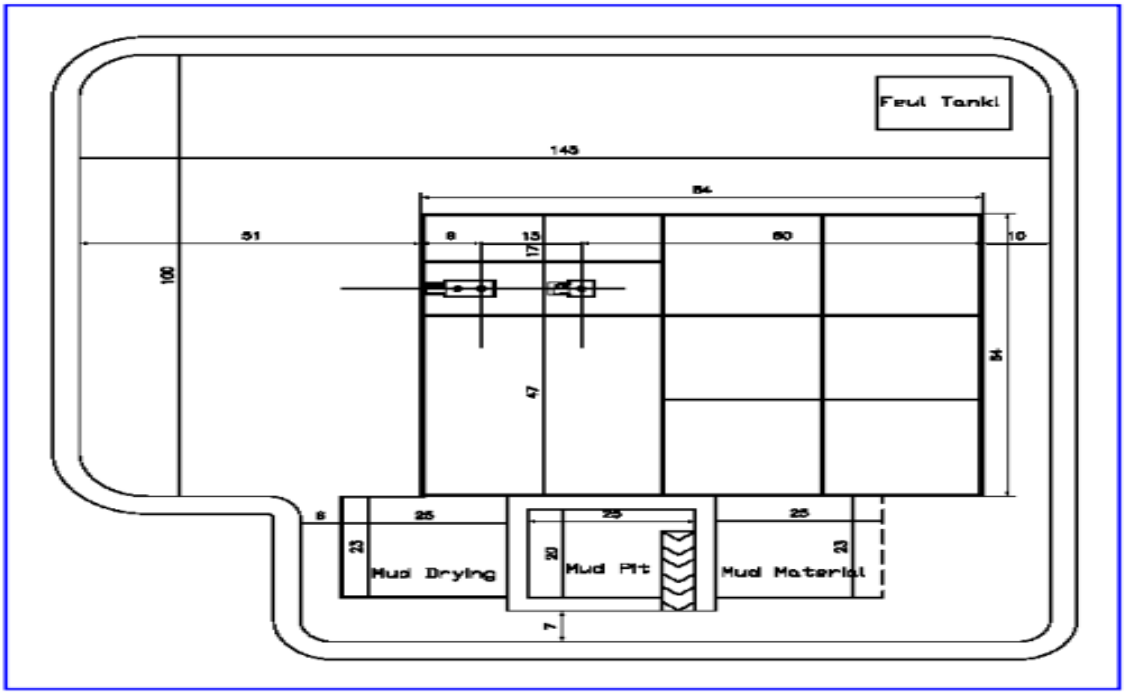
Table 3: Well Pad Specification

Compressive Strength for reinforced concrete after 28 days: 280 kg/cm2
• Fuel tank concrete pond dimension: Length 20m×Width 12m×Height 0.5m
• Drainage: Width 30cm × Depth 30cm
• Burning pit: 15m×15m square, located on South-East or South-West with at least 350m distance from well head
• Cellar to cellar distance in multi-well well pad: 15m
• Conductor pipe depth under cellar bottom is 5.5m. The conductor pipe will be driven while constructing the well pad.
• Mud material dimension is 25m*35m +granular surface
• Mud drying dimension is 25*35m + concrete surface
• Well pad surface is gravel. Access road granular surface. The level of well pads is 3.95 MSL

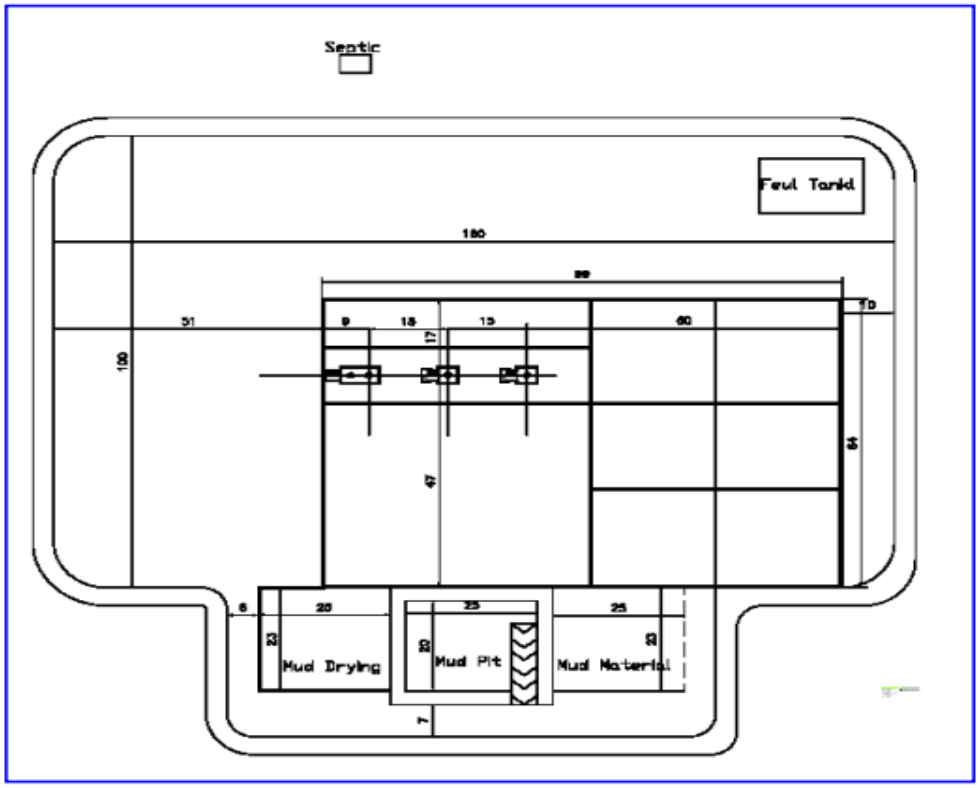
1. Types of well pads:
• One well in one pad.
• Two wells in one pad. 2 Sarvak horizontal wells or 1 Sarvak horizontal well plus 1 vertical/directional well).
• Three wells in one pad: 2 Sarvak horizontal wells and 1 vertical/directional wells.



Well Pad for one well



Double-well Well Pad



Triple-well Well Pad

Figure 3: Well Pads

1.5 Predicted Formations Top and Thickness

Table 4 illustrates the actual formation geological succession, top and thicknesses that penetrated in AZN-10 (All measured depth (MD) are based on RTE@ 13.45m)

Table 4: Actual Formation Top of AZN-10

Formation /Member Name		Top		Thickness
		mTVDss	mMD	mTVD
Aghajari		Surface	Surface	1142.0
Gachsaran		-1138.0	1151.5	350.5
Asmari		-1488.5	1502.1	282.7
Pabdeh		-1771.2	1784.8	415.0
Gurpi	U-Gurpi	-2186.2	2199.8	121.8
	Tarbur	-2308.0	2321.6	85.6
	L-Gurpi	-2393.6	2407.2	125.1
Ilam		-2518.7	2532.3	107.9
Laffan		-2626.6	2640.2	6.1
Sarvak		-2632.7	2646.3	652.0
Kazhdumi		-3284.7	3298.3	198.8
Dariyan		-3483.5	3497.1	197.1
Gadvan		-3680.6	3694.2	293.6
Fahliyan	Fahliyan	-3974.2	3987.8	37.6
	Fah-1a	-4011.8	4025.4	14.7
	Fah-1b	-4026.5	4040.1	15.7
	Fah-1c	-4042.2	4055.8	32.4
	Fah-2a	-4074.6	4088.2	15.3
	Fah-2b	-4089.9	4103.5	10.2
	Fah-3a	-4100.1	4113.7	10.0
	Fah-3b	-4110.1	4123.7	9.7
	Fah-4a	-4119.8	4133.4	10.5
	Fah-4b	-4130.3	4143.9	13.5
	Fah-5a	-4143.8	4157.4	21.4
	Fah-5b	-4165.2	4178.8	116.5
Garau		-4281.7	4295.3	480.3

Gotnia	-4762.0	4775.5	149.0
Najmeh	-4911.0	4924.5	26.0
Sargelu	-4937.0	4950.5	108.0
Alan	-5045.0	5058.5	40.0
Mus	-5085.0	5098.5	92.0
Neyriz	-5177.0	5190.5	
TD	**-5326.2**	**5340.0**	

1.6 Formation Description

A brief description of the Sarvak units are as following:

Fahliyan Formation

This formation is start with light gray, light green gray, soft Limestone interbedded with green gray, soft, sticky Marl, green gray, light gray, soft, Argillaceous Limestone and dark green gray, soft, sticky Clay stone beds. This formation at continue is consisting of buff white light gray, light brown, brown, medium hard, hard, densely, slightly argillaceous Limestone and off white, white gray, medium hard Chalky Limestone beds.

1.7 Predicted Pressure, Temperature and OWC

The predicted pressure, temperature and TTZ of reservoir AZN-10 based on available data are presented in Table 3. The Formation Pressure Prediction in Block II is illustrated in Table 4. The formation pressure prediction is the result of dynamic model.

Table 1: Fahliyan Reservoir Predicted Pressure, Temperature and TTZ Contact

Reservoir	Sub-layer	Mid of SubLayer Depth in AZN-10 (mss)	Temp (°F)	Temperature Gradient (°F/100ft)	Current Reservoir Pressure (psia)	Basis for Reservoir Pressure	Top of Transition Zone (mss)	Basis for TTZ and FWL
Fahliyan	Fah-1	-4043	260	1.32	9260	Dynamic Model	-4149	Dynamic Model
	Fah-2a	-4082	261		9328	Pressure Gradient	-4045.6	
	Fah-2b	-4095	262		9347	Pressure Gradient	-	
	Fah-3a	-4105	262		9183	Dynamic Model	-4151	
	Fah-3b	-4112	263		9195	Pressure Gradient	-	
	Fah-4a	-4125	263		7447	Dynamic Model	-4122	
	Fah-4b	-4137	264		7471	Pressure Gradient	-4107	
	Fah-5a	-4154	264		8111	Dynamic Model	-4186.5	
	Fah-5b	-4193	266		8241	Dynamic Model	-4186.5	

Formation	Interval	Pore Pressure	Collapse Pressure (Break-Out Pressure)	Fracture Pressure (Break-Down Pressure)
	m	*SG*	*SG*	*SG*
Aghajari	0～1151	0.98～1.18	0.06～1.39	1.01～1.89
Gachsaran	1151～1502	1.13～1.31	0.78～1.36	1.89～2.01
Asmari	1502～1785	1.14～1.26	0.6～1.28	1.43～1.79
Pabdeh	1785～2200	1.20～1.26	0. 82～1.36	1.40～1.78
Gurpi	2200～2532	1.21～1.26	0.66～1.02	1.40～1.76
Ilam	2532～2640	1.18～1.24	0.20～0.9	1.41～1.72
Laffan	2640～2646	1.13～1.18	0.24～1.11	1.41～1.72
Sarvak	2646～3298	1.14～1.26	0.24～1.23	1.44～1.81
Kazhdumi	3298～3497	1.1～1.14	0.46～0.96	1.93～1.94
Dariyan	3497～3694	1.18～1.25	0.55～1.10	1.96～2.06
Gadvan	3694～3987	1.1～1.25	0.55～1.44	1.97～2.11
Fahliyan	3987～TD	1.60～1.3	1.20～1.75	2.08～2.12

Table 6: Prediction Formation Pressure

1.8 Drilling History

The AZN-10 well was drilled from 25-June-2008 to 23-March-2010 to total depth 5340m and abandoned with cement plugs as describe in the following.

26" Hole Section Summary:

This section was drilled from surface to 152.5m with 26" bit & Mw 65pcf, cased with 20"(K-55, 133#) at 151m with 240bbl cement & S.W 115pcf, cement return observed, well equipped by 20-3/4" – 3000psi Wellhead & related BOP & Acc.

17-1/2" Hole Section Summary:

This section was drilled with 17-1/2" bit to 1502m with gradual increase of Mw f/74 to 90pcf because of tight hole, pressure drop observed several times & a fishing Op. with overshot at 1264m, took CR/SONIC/CALLIPER/CDR logs, cased with 13-3/8" (N-80, 72#, 8RD&BTC) at 1502m with S.W 115pcf, return cement observed, well equipped by 13-5/8" – 5000 psi Wellhead & related BOP & Acc.

12" Hole Section Summary:

This section was drilled with 12" bit to 2731m with 0-8BPH losses and Mw 85pcf, waiting due to repair from GWDC Company, mud loss observed amount of 45 & 50BPH at 2690 & 2731m, loss controlled with LCM Pill, continue drilling to 3002m with 68BPH losses, condition trip & took GR/CNT/LDT/HNGS/BHC/HRLA/MSFL logs in two runs, continue drilling to 3857m with 0-6BPH losses and gradual decrease of Mw f/85 to 84pcf, GR/WSTTCDR and SDL/DTD/GTET/CSNG/DSN and GR/MSFL/DLL were taken in 3 three runs, 9-5/8" casing (L-80, T-95, 53.5#, SEC) was run to 3854m, cemented with S.W 95 & 115pcf, Cement return observed, well equipped by 11" – 10,000psi Wellhead & related BOP & Acc.

8-3/8" Hole Section Summary:

Displaced well with 100pcf mud and continue drilling to 3867m with increasing Mw to 105pcf, continue drilling to 4073m with increasing Mw to 110pcf and 2-4% gas recorded f/4054 to 4062m, continue drilling to 4279m with 1-3BPH losses, RIH with Junk Sub, work on junk & POOH continue drilling to 4399m with 5BPH losses & 14% gas recorded, CDR/GRMSFL/DLL & CSNG/DSNT/SDL & CSNG/DSNT/SDL & SONIC/GR were taken in two runs, condition the hole & took MDT log & check pressure condition in 30 points from 4025 to 4221m, L/D top drive due to lack of efficiency and its rotating system & P/U Kelly, waiting due to TDS repair & providing Kelly spinner from GWDC Company, continue drilling to 4473 with 1-3BPH losses, stuck pipe at 4430m & 4439m and get free with 80-100000 over pull, while RIH after changing bit at 270m observe a part of BHA dropped in hole, catch fish & POOH to surface, continue drilling to 4536m with 1-2BPH losses and 1-7% gas recorded, pipe stuck several times & gradual Mw increasing to 115pcf, pipe stuck at 4536 and 90% gas was observed with return mud & 10 bbl. increase in tank volume, control the well with 128pcf Mw, continue drilling to

4925.5m with 2-4BPH losses and record 20-80% gas while drilling and increasing Mw to 131pcf, losses controlled with LCM Pill & observed 3BPH flow, control the well with 135pcf Mw while wash & ream hole in some intervals, stuck pipes and free themes at 4890 m , took MSFL/DLL/GR & SDL/DSN/TLD/GPIT logs in two runs, condition trip & unsuccessful run GR/CSNG/DSN/SDL logging tools to 3880, condition the hole unsuccessful run DSI/GR logging tools to 3970m, stuck and free tools, hole cleaning with bit & reamer, run NGT/CNT/TLD/GPIT logging tools to 4913m & took log to 4400m, condition trip & run 7" liner (T-95, MW95, SSHC, 35#, BDS) f/3707.5 to 4922m, (Liner lap at 3707.5 m) Liner was cemented (S.W 138pcf). Scrap to liner lap and perform pressure test & dry test was ok.

5-7/8" Hole Section Summary:

RIH with 5-7/8" bit, drill out POB & plugs, dry test with 2000psi in 15min, was ok. Mw balanced to 120pcf and drilling to 4940.5m with 0-2BPH losses, Mw decrease f/120 to 118pcf, 30BPH losses controlled with LCM Pill and decreasing Mw to110pcf, continue drilling to 5068 m, observe tight hole & increasing Mw to 113pcf, continue drilling to 5230m with 0-3BPH,Loss return control with LCM Pill and continue drilling to 5230m with 2-3 BPH losses with tight hole in several points, took LDL/CNL/NGS logs at 5212 m (stop in this point) to shoe & HRLT/GR at 5132 to shoe in two runs, RIH with PDC bit to 5230m (TD) with reaming & hole cleaning f/4987, stuck pipe at 4989m & 5002m got free with 80000lb over pull without jarring, meanwhile Mw increase f/113 to 114pcf, 70BPH losses at 5108 & Circ. LCM Pill,0-2BPH losses, continue drilling to 5315 with 0-10BPH losses, Cir. LCM Pill, tight hole & extra tension & pressure drop from bit and jar, stop drilling due to logging, POOH meanwhile pumping and stuck pipe at 5285m,work on stuck without result, electrical back-off at 5191m, pipe get free, POOH to surface, fish left in hole 94.68m including 10 joints HWDP 4-3/4", Bit Sub, stabilizer, Bit and XOS fish top at 5190.32m

F.B D.S.T Test No. 1:

Repeatedly scraping f/4848 to 4902m & spotting high vis. Pill, run F.B D.S.T equipment (filling with water as a cushion), packer set at 4882.2m, performed F.B D.S.T, mud reverse Circ., packer get free, POOH & L/D F.B D.S.T.

F.B D.S.T Test No. 2:

RIH with bit due to perform hole condition, RIH with open ended pipe 2-7/8" due to put balance cement plug at 5185m in two runs, get weight at 5118 in second run, drilling cement to 5116m, scraping f/4850 to 4900m, spot high vis. Pill, run F.B D.S.T (filling with water as a cushion), packer set at 4872.6m, perform F.B D.S.T, unsuccessful to open reverse Circ. Valve, packer get free, POOH to surface, L/D F.B D.S.T, release gas pressure trapped in tools, slow operation due to existing H_2S in pipes.

F.B D.S.T Test No. 3:

Hole condition with bit, RIH with open ended pipes, put cement plug at 5055m, wait on cement to get hard, tag cement at 4996, repeatedly scraping f/4800 o 4890m, spot high vis. Pill, run F.B D.S.T (filling with water cushion), packer set at 4864m, RIH CT to 4830m, open well by STV, continue RIH CT along with acidizing, POOH CT to surface, inject acid with diesel, observe water & diesel from flow line, mud reverse Circ., L/D CT, packer get free, POOH to surface, leak observed from SSV, L/D F.B D.S.T.

Performed F.B D.S.T Test No. 4

Meanwhile prepare F.B.DST tolls, RIH to 4996 m (P.B), spot high vis. Pill, RIH F.B.DST (fill up with water as a cushion water), set packer at 4861 m and perform F.B.DST, reverse circulation, equalize MW, try to release packer with 60-100000 lb. without jar no success, rig up and RIH electrical back off string to 4779 m, perform electrical back off POOH electrical back off String and L/D,POOH to surface , part of R.D S.V mandrel with accessories with length of 26.29 m left in hole, top of fish at 4837.75 m, RIH with part of R.D S.V tolls and jar to 4837.75 m, touch fish and catch it, work on releasing packer with jar and 80000 lb. max over pull, fish get free, recover fish, L/D F.B.DST.

5-7/8" Directional Drilling Summary:

Drilling cement f/4996 to 5170m with 0-1BPH losses, prepare 1200 bbl. OBM displace the well with 92 pcf. OBM, Circ. Mud & Mw increasing to 95pcf, POOH to surface, trying to tilt the well in 4 runs with Open Hole Whip Stock 5-7/8" tool in several depth without any results from date 25-July-2009 to 3-Oct-2009, RIH with open ended pipes, put cement plug as balance at 5060m,wait on cement to get hard, tag cement at 4964m, POOH to surface, drilling cement to 4990m, put 10,000lb weight on cement, RIH with 5-7/8" bit K300 with Bent Sub (2°) & downhole motor 4-3/4" to 4990m, pipes were weighed, start tilting the well to 5010m with 0-2BPH losses (KOP @ 5000m), stop drilling, POOH to surface, RIH with new bit & Near Bit Sub & continue drilling f/5010 to TD (@ 5340m) with 0-3BPH losses & changing STB position in several times with 1-4BPH losses and 1-4% gas detected in mud (f/5104.5 to 5303), stop drilling due to take log, CBU, 1.5% gas recorded by mud logging sensor, run HRI/CDR/GR logging tools to bottom, start logging to 4950m (CDR connection failed) & truck system couldn't display the log, POOH & L/D logging tools, took CDR/SDL/DSN/CSNG & BSC/GR/CCL from bottom of hole to shoe & CBL/VDL/CCL/GR logs to 3700m, RIH to bottom with good hole condition, run VSP logging tool to 5063m which didn't go down due to high Temp and Working Tem. limitation, take logs totally in 224 points f/5063 to 260m, hole conditioning with bit & stabilizer, Rig up casing running equipment, 5" liner (T-95, VAM, 18#) run and cemented to 5193.5m (LDC @5169m & Lap @4781m), perform dry test, RIH with J.Mill & drilling POB, unsuccessful LDC drilling & cement below with J.Mill in two step, RIH with Junk Sub, work on junk, & drilling LDC with Mill & downhole motor with reaming & hole conditioning trip to bottom, no change in mud volume, POOH to 4780m & spot high vis. Pill, POOH to surface.

F.B D.S.T Test No. 5:

RIH with 7" F.B D.S.T (filling with water as a cushion), packer set at 4751.7m, perform test, put 1400psi pressure inside annulus to open RD which suddenly drop to 0, pumping 13bbl mud into coiled tubing, increasing pressure observed inside tubes to 3300psi, increase pressure to 4000 psi with pumping 60 bbl , observe connection between completion string and annulus, mud reverse Circ., with burning inner string gas in burner line, release packer with 80,000lb over-pull, POOH to surface & L/D F.B D.S.T, operation stopped due to casing wrench failure & air compressor generator (No.4) get fire, disconnect power, try to take CBL/VDL logs in two runs (second run with hole finder) which didn't go lower than 4793m, RIH in 2 trips (first trip with 4-1/8" bit RR & 5" scraper, second trip with J.Mill 5-7/8") which in first trip didn't go lower than 4780m & in second trip didn't go lower than 4779.5m, hair-shape iron filings observed on shakers in second trip & erosion marks on J.Mill, RIH with Taper Mill 4-1/8" to 4781.5m & took weigh, try to pass depth which didn't go lower than 4782.4m, POOH to surface due to lack of progress, RIH with previous J.Mill 4" to 4782m & took weight, try to pass the depth with torsional torque fluctuation, string stopped at 4782.3m, continue RIH to 5183m & took weight, RIH with rotation & pumping to 5203m, wash and ream f/5203 to 5264m with no change in mud volume, stuck pipe at 5236m while reaming, work on freeing pipes with spotting 50bbl Pipe Lax, in two parts with decreasing Mw f/112 to 103pcf, pipes get free with torque transfer & 130,000lb over-pull, POOH to surface, hole condition trip with bit, RIH with open ended pipes, put cement plug as balance at 5337m, in 3 steps(7 bbl in first and second step and 12 bbl in third step), tag cement at third step at 5264m, unsuccessful wash to 5264.8m with 5000lb weight, POOH to surface, RIH with J.Mill 4-1/8", unsuccessful wash and ream f/5202 to 5260m with pumping & 6000lb weight, POOH to surface, scraping f/4600 to 4775m.

F.B D.S.T Test No. 6:

RIH with 7" F.B D.S.T (fill-up with water as cushions), packer set at 4741m, perform test, mud reverse Circ., unset packer, 9BPH losses, direct Circ. & POOH to surface, with zero losses, L/D F.B D.S.T tools, RIH with 2-7/8" open ended pipe, not going down from 5201 m, condition hole with junk mill, RIH with 2-7/8" open ended pipe put cement plug as balance at 5260m (13 bbl), tag cement at 5191m, POOH to surface, drilling cement with 4-1/8" bit to 5230m, POOH to surface, scraping f/4600 to 4770m, spot high vis. Pill.

F.B D.S.T Test No. 7:

RIH with F.B D.S.T ((fill-up with water as cushions), packer set at 4732m, perform test, mud reverse Circ., free packer, 4BPH losses, POOH to surface with 0-2BPH losses, L/D F.B D.S.T tools, RIH with 4-1/8" bit & 5" scraper to 5170m (pass liner lap with 1/2 rotating & 1000-2000lb weight), repeatedly scraping f/4782 to 5170m, spot high vis. Pill, POOH to surface while spotting 30bbl 80pcf mud at 4770m, unsuccessful to take GR/CBL/VDL logs in 2 runs due to tools failure, take GR/CCL log f/5190 to 4725m, run Gauge Ring/Junk basket/CCL tools to 5160m, run & set 5" Bridge plug at 5147.5m, perform test with 3000psi in 15min was good., L/D extra pipe on rig.

Performed F.B D.S.T Test No. 8

RIH CNLC perforation tolls with failure of depth Encoder before start operation, change it, perforate f/4959-4968 m, RIH F.B.DST (fill-up with water as a cushions), set packer at 4721.5 m and perform F.B.DST, wait on NIDC Stimulation in two steps, reverse circulate, unset packer, POOH to surface with 0-3 BPH loss, lay down F.B.DST.

Secure & Suspend:

RIH with open ended pipes, put CMT plug at 4780m (18bbl & 118pcf), while wait on cement, RIH in 2 steps to 4638m, mud Circ., dry cement observed on shakers, POOH to surface & RIH with 5-7/8" bit & 7" scraper to 4636m & get weight, drilling hard cement to 4663m & POOH to surface, performed cement dry test, RIH with open ended pipes & put CMT plug at 3707m (30bbl & weight 115pcf) while wait on cement, RIH in 2 steps to 3620m without touching cement, folliculated cement observed on shakers in two steps, POOH to surface, RIH with 8-3/8" bit, drilling cement f/3618 to 3630m, POOH to surface, RIH with open ended pipes to 1502m & put CMT plug (35bbl & 115pcf),meanwhile wait on cement, POOH to surface, RIH with 8-3/8" bit & drilling hard cement f/1360 to 1378m, POOH to surface & L/D drill pipes. N/D BOPs & install Blind Flange 11" – 10,000 psi, perform test with 4,000psi for 15min was ok, observe annulus was ok. Open extra valves of spools, install wellhead Acc., secure the well, rig was released at 23-March-2010.

1.9 Casing & Liner Depth

Casing Spec.	20"	K-55, 133#	0-151 m
	13-3/8"	N-80, 72#	0-1502 m
	9-5/8"	T-95,53.5#	0-3854 m
	7"	T-95, 32#	3707.5 – 4922 m
	5"	T-95, 18#	4781 – 5193.5 m

Table 7: Casing & Liner Data

1.10 Well Schematic before Workover

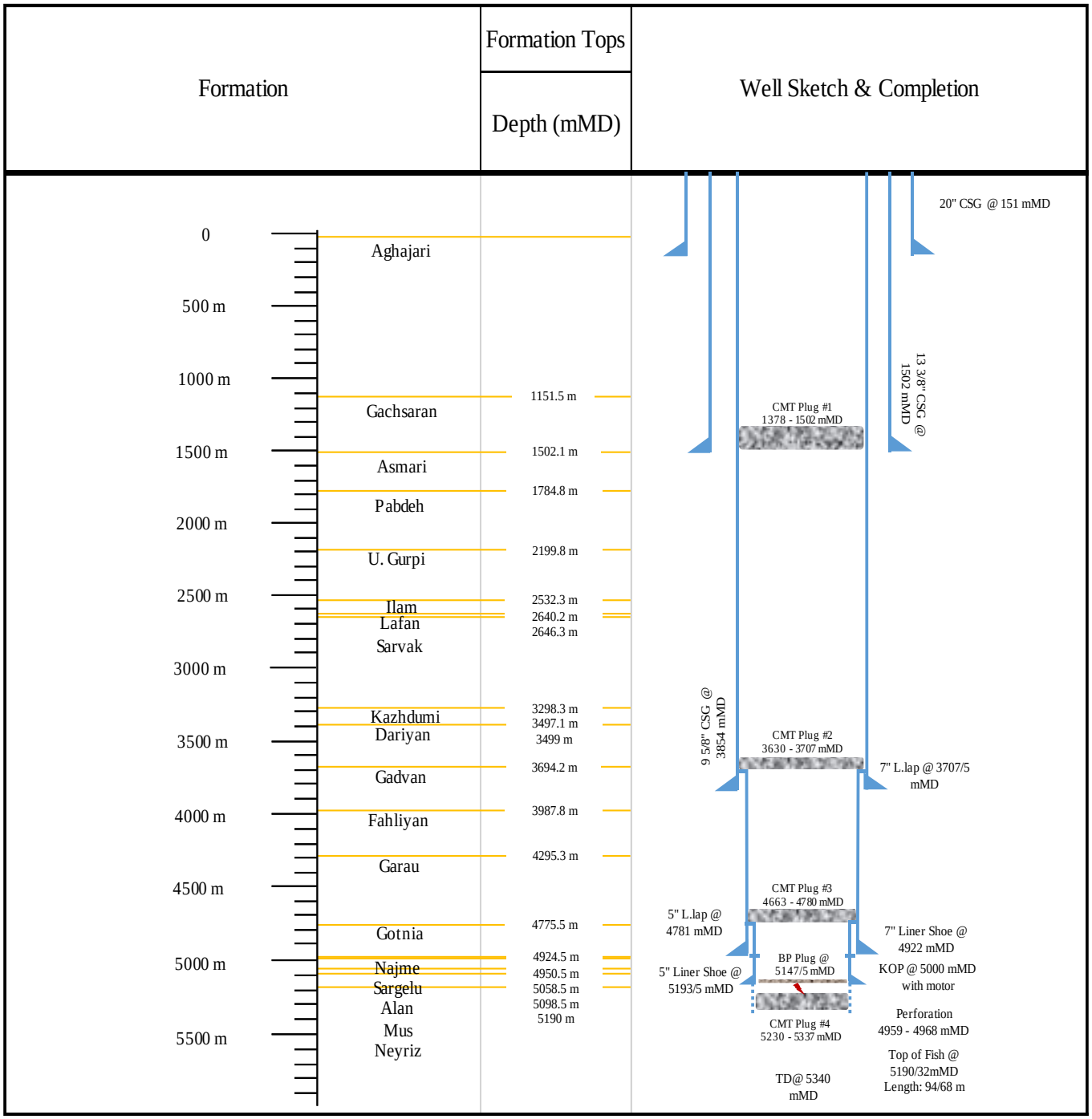


Figure 4: Well Schematic before workover

1.11 Well Schematic after Workover

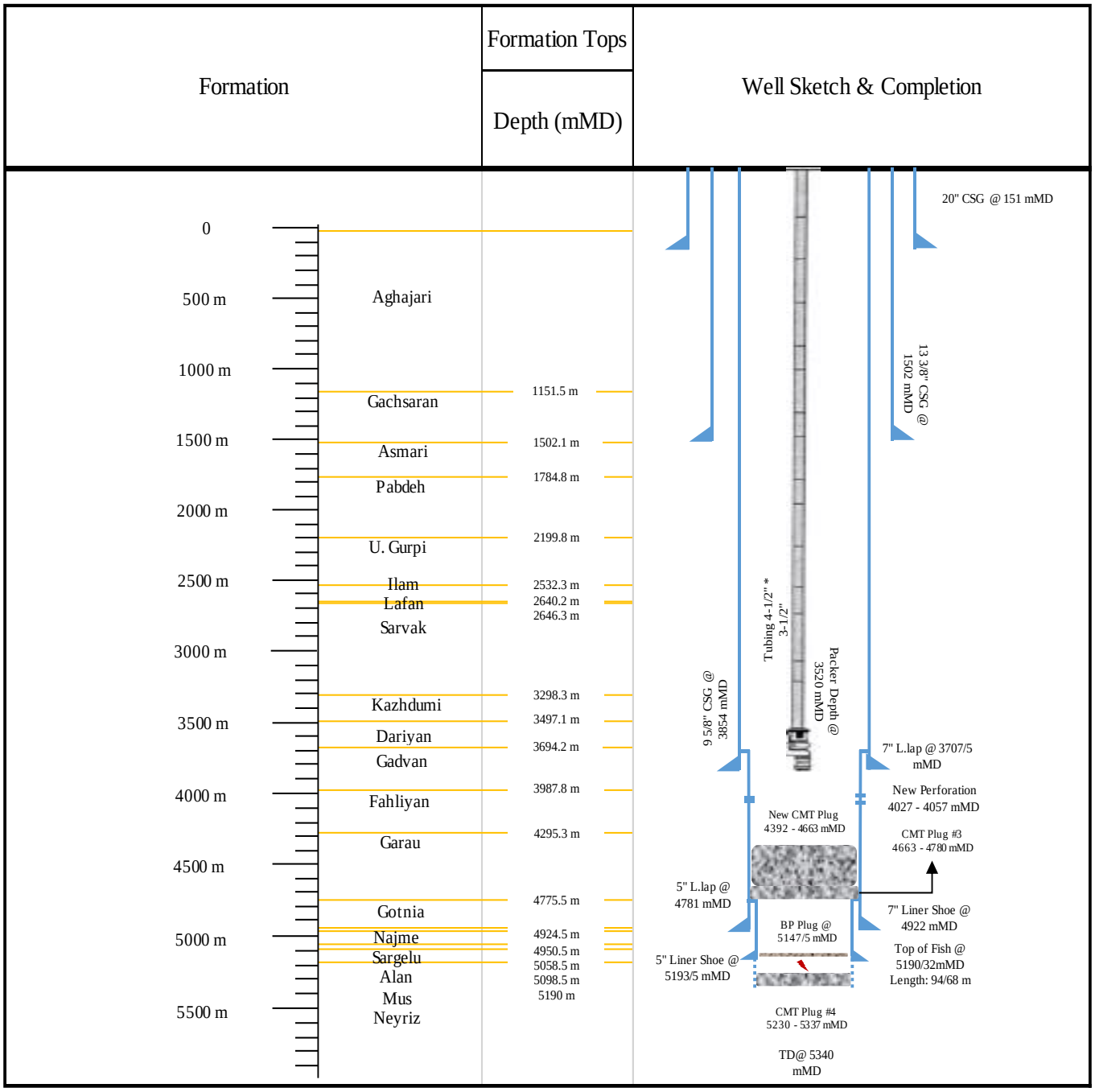


Figure 5: Well Schematic after workover

2. Geological Information

2.1 Formation Stratigraphy and Lithology

Stage	Group	Formation	Lithology	Lithology Description	Block I Top (mTVD)	Block II Top (mTVD)	Block III Top (mTVD)
Pliocene Miocene	Fars	Agha Jari		Marl, Sandstone, Anhydrite	Surface	Surface	Surface
		Gachsaran		Marl, Anhydrite, Claystone	1380	1150	961
Miocene							
E. Miocene Oligocene		Asmari		Limestone, Sandstone, Claystone	1776	1503	1318
E. Oligocene L. Paleocene		Pabdeh		Limestone, Argillaceous Limestone, Claystone, Dolomite	2079	1786	1525
Maastrichtian Campanian		U. Gurpi Tarbur L. Gurpi		Limestone, Argillaceous Limestone	2352	2303	2153
Santonian Coniacian	Bangestan	Ilam		Limestone, Claystone	2606	2534	2549
		Laffan		Claystone	2691	2642	2625
Turonian Cenomanian		Sarvak		Limestone, Claystone	2697	2648	2632
Cenomanian Albian		Kazhdumi		Sandstone, Argillaceous Limestone, Claystone, Shale	3335	3300	3269
Aptian	Khami	Dariyan		Limestone	3538	3499	3471
Aptian Barremian		Gadvan		Argillaceous Limestone, Claystone, Sandstone	3718	3696	3635
Neocomian		Fahliyan		Limestone, Marl	4092	3989	3974

Figure 6: Azadegan Stratigraphic Column

3. Logging Program

The logging program has been tabulated in Table 6, which shall be performed by Contractor in cased-hole, section by section.

Table 8: Proposed Logging Plan forAZN-10 (Near Pad-50)

Hole Size	Casing / Liner	Cased Hole Wire Line Logging
12-1/4"	9-5/8"	Corrosion Log
8 1/2"	7" Liner	1. CAST/CBL/VDL/CCLGR or SBT/VDL/CCL/GR Or USIT/CBL/VDL/CCL/GR 2. Corrosion Log 3.CCL/GR

4. Program Outline

4.1 Contractors Involved

Table 9: Contractor Involved

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC 201
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Hafaran Arvand
4	Electrical Logging and Perforation	PANIZ
5	Mud Logging	MAPSA KISH
6	H2S Monitoring and Control	MAPSA KISH BONYAN IMEN
7	Casing and Tubing Running	OEOC
8	Cementing	SPAD ENERGY
9	Dry Test, DST and RTTS	OEOC
10	Slick Line	OEOC
11	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
12	Surface Well Testing	OEOC
13	Liner Hanger	PARS POOLAD
14	Well Head	OEOC

4.2 Time Breakdown

De-Complete, Work-over and Completion Operation TBD of Well: AZNS-10 Work-over	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Work-over									
Held PJSM. Spud the Rig. Stand Back required 5" DPs, 5" HWDPs and 6-1/2" DC.	SPUD	5340	0.0	0.00	0.0	0.0	0.0	0.00	0.00
Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & Bleed off any annulus pressure to Zero upon client's approval and recheck the well status. Note: If annulus pressure is not Zero after bleed off, follow contingency plan #1.	Wellhead	5340	4.0	0.17	0.2	0.2	4.4	0.18	0.2
Remove 11" blind flange & change wellhead section C with new one.	Wellhead	5340	18	0.75	0.9	0.8	19.8	0.83	1.0
N/U BOP and pressure test. Perform wellhead pressure integrity test.	BOP	5340	14	0.58	1.5	0.6	15.4	0.64	1.7
RIH with 8-1/2" BHA & drill out cement plug #1 F/1378m to 1502m.Continue drill out cement plug #2 F/3630m to 3707m, tag top of 7" liner, circulate till shakers clean & POOH to surface.	D/O	3707.5	84	3.50	5.0	3.9	92.4	3.85	5.5
RIH with 5-7/8" Bit & tag cement plug #3 @ 4663m, Dress the CMT (drill out soft cement if any), CBU till shakers clean, POOH & L/D Bit.	D/O	4663	48	2.00	7.0	2.2	52.8	2.20	7.7
RIH with 2-7/8" open end DPs, put CMT plug from 4663m to top of Garau (base of Fahliyan, 4392m), POOH to surface.	Plug Back	4392	72	3.00	10.0	3.3	79.2	3.30	11.0
RIH with 7" & 9-5/8" tandem scrapper & scrap inside 9-5/8" (RTTS PKR depth, 3620 - 3680m) & 7" liner, POOH to surface.	Trip	4392	48	2.00	12.0	2.2	52.8	2.20	13.2
Perform BOP test.	BOP Test	4392	6	0.25	12.3	0.3	6.6	0.28	13.5
RIH with wireline logging tools (SBT or CAST/CBL/VDL) inside 7" liner, perform logging, POOH.	Log	4392	24	1.00	13.3	1.1	26.4	1.10	14.6
RIH with wireline corrosion logging tool	Log	4392	24	1.00	14.3	1.1	26.4	1.10	15.7
RIH RTTS PKR & perform 7" liner lap & cement plug dry test, POOH to surface. Note: If dry test is not ok, follow contingency plan #2.	Trip	4392	42	1.75	16.0	1.9	46.2	1.93	17.6
RIH with 3-1/2" DP to 4392m (top of the new CMT plug), displace the 7" liner with completion fluid (CaBr2)	Completion	4392	38	1.58	17.6	1.7	41.8	1.74	19.3
Perform BOP test.	BOP Test	4392	6	0.25	17.8	0.3	6.6	0.28	19.6
Run perforation gun assembly & perforate 7" liner by TCP (F/4027m to 4057m, Fah-1a & 1b), POOH to surface	Completion	4392	60	2.50	20.3	2.8	66.0	2.75	22.4
Pick up 5 7/8" Mill & RIH W/ Clean out BHA, Clean Inside 7" Liner to TD & displace the hole with CaBr2, POOH while L/D DP's.	Completion	4392	40	1.67	22.0	1.8	44.0	1.83	24.2
RIH with GR-CCL log o confirm perforation intervals.	Completion	4392	12	0.50	22.5	0.6	13.2	0.55	24.8
RIH with DP's & DC's, POOH while L/D same.	Completion	4392	48	2.00	24.5	2.2	52.8	2.20	27.0
Perform BOP test.	BOP Test	4392	6	0.25	24.8	0.3	6.6	0.28	27.2
Run 10K completion string as per program.	Completion	4392	96	4.00	28.8	4.4	105.6	4.40	31.6

Set BPV inside tubing hanger, N/D BOPs.	Completion	4392	14	0.58	29.3	0.6	15.4	0.64	32.3
Retrieve BPV, install TWCV in TBG Hanger,N/U X-mass Tree,Perform Required test as per program,Retrive TWCV	Completion	4392	8	0.33	29.7	0.4	8.8	0.37	32.6
R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line. retrieve plug & prong from NO GO landing nipple, POOH.	Completion	4392	8	0.33	30.0	0.4	8.8	0.37	33.0
R/U CTU, Stimulation and Well Testing Eq. RIH w/ CT to Btm, displace the well with Diesel, POOH CT to surface.	Completion	4392	24	1.00	31.0	1.1	26.4	1.10	34.1
R/U Slick Line, Retrieve protection sleeve from TRSSSV, R/D Slick Line.	Completion	4392	4	0.17	31.2	0.2	4.4	0.18	34.3
Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion	4392	80	3.33	34.5	3.7	88.0	3.67	38.0
RIH with S/L, set protection Sleeve, POOH & R/D SL.	Completion	4392	Rig Less	0.00	34.5	0.0	0.0	0.00	38.0
RIH with C/T & perform acid wash as per program, POOH to surface.	Completion	4392	Rig Less	0.00	34.5	0.0	0.0	0.00	38.0
RIH with S/L, retrieve protection sleeve, POOH & R/D SL.	Completion	4392	Rig Less	0.00	34.5	0.0	0.0	0.00	38.0
Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs. NOTE: The ime is only for flowing duration (R/U, R/D & preparation time is excluded.)	Completion	4392	Rig Less	0.00	34.5	0.0	0.0	0.00	38.0
RIH with S/L, set protection sleeve.	Completion	4392	Rig Less	0.00	34.5	0.0	0.0	0.00	38.0
RIH with C/T to bottom & displace the well with diesel, POOH.	Completion	4392	Rig Less	0.00	34.5	0.0	0.0	0.00	38.0
RIH with S/L, install plug & prong into NO GO landing nipple, POOH.	Completion	4392	Rig Less	0.00	34.5	0.0	0.0	0.00	38.0
RIH with S/L, retrieve protection sleeve.	Completion	4392	Rig Less	0.00	34.5	0.0	0.0	0.00	38.0
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, Release the Rig.	Completion	4392	Rig Less	0.00	34.5	0.0	0.0	0.00	38.0
Total Duration of Work-over:				***33.0***	***34.5***			***38.0***	***38.0***
NOTE:									
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip	4392	4	0.17	34.7	0.2	4.4	0.18	38.1
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service	4392	19	0.79	35.5	0.9	20.9	0.87	39.0
Total Duration of AZNS-010 Work-over Well:		**4392**			**35.5**				**39.0**
Contingency Plans TBD of Well: **AZNS-010 Work-over**	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Based on operation situation in each step, the following contingency plans might be followed									
Contingency plan #1: Release pressure and monitor the pressure. If wellhead pressure is not zero, contractor shall inform client. Remidial plan to be prepared in advance by operators and will be followed after client's confirmation. Note: Time will be confirmed by client's approval.	Wellhead	5340	TBA	TBA	TBA	TBA	TBA	TBA	TBA
Contingency plan #2: RIH W/7" RTTS Packer and perform dry test for the cement plug. Note: If dry test result in contingency plan #2 is ok, so liner lap is leaking. Thus follow contingency plan #2-1 as following.	Trip	4392	98.4	4.1	39.6	4.51	108.24	4.51	4.5
Contingency plan #2-1: RIH and install liner top packer on top of 7" liner lap. POOH	Trip	4392	98.4	4.1	43.7	4.51	108.24	4.51	9.0
Total Duration of AZNS-010 Work-over Well:		**4392**			**8.2**				**9.0**

5. Sequence of Operation

5.1 Main Objective

Based on test results the decision has been made to complete the well for Fahliyan production. In order to achieve these, following steps will be performed.

1. Move and spud the rig over well.

2. Check & record all wellhead pressures.

Note #1: If wellhead pressure is not zero, follow the contingency plan

Contingency plan: Release pressure and monitor the pressure, if wellhead pressure is not zero, the contractor shall inform the client. The remedial plan will be prepare in advance by operators and followed after the client's confirmation.

Note #3: Ensure that adequate fishing tools are available.

Note #4: Enough mud material, LCM & cement should be available.

Note #5: Ensure that all drilling tools & equipment are on well site & are serviceable condition.

3. Remove 11" blind flange and change wellhead section C with new one.

4. Nipple up BOPs set up 13-5/8" plug tester and pressure test same. Perform wellhead pressure integrity test with 9-5/8" cup tester.

Perform BOP test with 300/7500 psi for 5/15 min.

Pressure test shall be performed with both low pressure test, usually 300 psi and high pressure test, usually to the rated value 6500 psi for wellhead. All high pressure tests should be maintained for 10-30 minutes, and low pressure test 5 minutes.

Note #6: When casing cup tester is used, the above test pressure should not exceed 80% of the internal yield pressure of the exposed casing. Wellhead side outlets below the test tool must be open during a test to avoid the risk of pressure increase below the test tool if there is a leak.

5. Displace oil base mud with 106 pcf kill mud.

6. RIH with 8-1/2" BHA & D/O cement plug#1(1378-1502m), continue D/O cement plug#2 (3630-3707m). Tag 7" liner lap. Circulate till shakers clean, POOH to surface.

Note #7: D/O last 10m of first cement plug by closed system and circulation through the chock manifold.

7. RIH W/ 5-7/8" Bit and tag cement plug#3 at 4663m, D/O cement plug #3, POOH to surface & L/D bit.

8. RIH with 5-7/8" Center Cut Bit, put CMT plug from 4663m to top of Garau (base of Fahliyan, 4392m), tag top of cement plug @ 4392m, Wash Down to Get Hard Cement POOH to surface.

9. RIH 7" & 9-5/8" tandem scraper to 4392m & scrap inside 9-5/8" CSG & 7" liner, CBU, POOH to surface.

10. RIH with wireline logging tools (GR/CCLCBL/VDL/SBT or CAST or USIT) inside 7" liner, perform logging, POOH.
11. RIH with wireline corrosion logging tools o bottom, log up to surface.
12. RIH RTTS PKR and perform 7" liner lap and cement plug dry test, POOH to surface.
Note #8: if dry test is not okay, the contingency plan is RIH W/ 7" RTTS packer and perform dry test for the cement plug, If was ok it, so liner lap is leaking. Install liner top packer on top of 7" liner lap. POOH.
Note #9: Contingency Plan Time is not considered in this program time duration.
13. RIH with 3-1/2" DP's to 4392m, displace the 7" liner with completion fluid. ($CaBr_2$).
14. Run perforation gun assembly & perforate 7" liner by TCP Fah-1a, Fah-1b, Fah-1c with a thickness of 14.7m, 15.7m and 32.4m TVD respectively), POOH to surface.
Note #10: Exact perforation interval will be specified based on log result.
15. Pick up 5 7/8" Mill & RIH W/ Clean out BHA, Clean Inside 7" Liner to TD & displace the hole with $CaBr_2$, POOH while L/D DP's.
16. RIH with GR/CCL log to confirm perforation intervals.
Note #11: Will be checked to 10m below the perforation intervals.
17. RIH with DP's & DC's, POOH while L/D same.
18. Run 10K completion string as per program.
19. Set BPV inside tubing hanger, N/D BOP & install Xmass tree.
20. Retrieve BPV, install TWCV in tubing hanger, N/U Xmass tree, perform required test as per program.
21. RIH with S/L, set protection sleeve in TRSSSV, R/D S/L.
22. Stimulation & well testing Eq. RIH with C/T to top of perforation, displace the well with diesel or Nitrogen, POOH.
23. RIH with S/L, retrieve protection sleeve, POOH & R/D S/L.
24. Flow the well as pre-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.
25. RIH with S/L, set protection sleeve, POOH & R/D S/L.
26. RIH with C/T & perform acid wash as per program, POOH to surface.
27. RIH with S/L, retrieve protection sleeve, POOH & R/D S/L.
28. Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for minimum 96 hrs.
Note #12: The time is only for flowing duration (R/U, R/D & preparation time is excluded)
29. RIH with S/L, set protection sleeve.
30. RIH with C/T to bottom & displace the well with diesel, POOH.

31. RIH with S/L, install plug & prong into NO GO landing nipple, POOH.

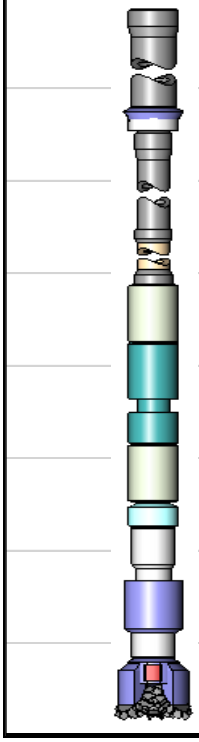
32. RIH with S/L, retrieve protection Sleeve.

33. Install BPV inside tubing hanger, secure the well & close all valves of Xmass tree expect BMV, release the rig.

6. BHA Program

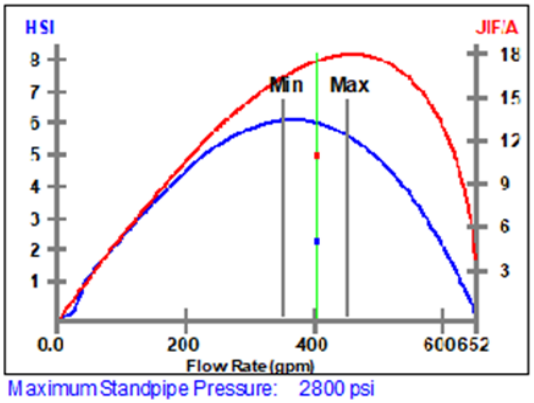
6.1 8-1/2" BHA

BIT SIZE (inch)	8-1/2"	**BHA Configuration to D/Out Inside 9-5/8" Casing**	**Collar**	**OD**
Depth In/Out (m)	1378 - 3707		DC-1	6-1/2"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	106			
B.FACTOR	0/784			
Weight in Air (lb)	24,624			
Bouyed Weight (lb)	19,297			
Bouyed Weight Below Jar (lb)	14,267			

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4/276	4-1/2" IF Pin x Box	-	
	5" Heavy Weight Drill Pipe	3	5	3	4-1/2" IF Pin x Box	27/42	182/71
	XOS	1	6-1/2	3	4-1/2" Reg. P x 4-1/2" IF B	1	1
	6-1/2" Drill Collar	3	6-1/2	2-13/16	4-1/2" Reg. P x B	27/42	155/29
	6-1/2" Jar	1	6-1/2	2.75	4-1/2" Reg. P x B	8	127/87
	6-1/2" Drill Collar	13	6-1/2	2-13/16	4-1/2" Reg. P x B	118/82	119/87
	6-1/2" Bit Sub W/ Float Valve	1	6-3/4"	3	4-1/2" Reg. B x B	0/75	1/05
	8-1/2" MT Bit	1	8-1/2"	-	3-1/2" Reg. P x B	0/3	0/3

6.2 8-1/2" Hydraulic Optimization

Hydraulics Optimization Bit Run Report



Maximum Standpipe Pressure: 2800 psi

Well Information

Well Name:
AZNS-10
Drilling Contractor:
MSA-OEOC OEOC-208
Operator:
MSA
Prepared For:

Location:
Near Pad 50
Prepared By:
Engineering Department/MSA

Hydraulic Details @ 3707 m

Flow Rate / Hole Diameter	47.1 gpm/in
System Pressure Loss	1306 psi
Bit Nozzle Pressure Drop	602 psi
Annular Pressure Drop	347 psi
Pump Input Hydraulic Power (M.E.=85%)	445 hp
Bit Hydraulic Horsepower	140 hp
Jet Impact Force	639 lbs
Impact Force/ Hole Area	11.26 lbs/in²
Hydrostatic Head	8944 psi
Equivalent Circulating Density	14.72 ppg
Cuttings Slip Velocity (Chip Size = .30 in)	10.45 m/min
Surface Equipment Length	100 m X 76.2
Buoyant Collar Weight	39550 lbs

Bit Run Data

Bit Number	1
Bit Size	8 1/2 in
Bit Type	517

Recommended Hydraulics

Nozzles	16\16\16 32nds
Total Flow Area	0.58 in²

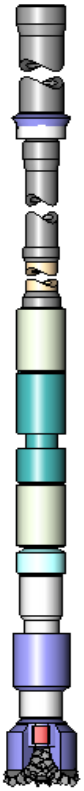
	IN	OUT
BIT HSI	2.47	2.47
JIF/Area	11.27	11.26 lbs/in²
Nozzle Velocity	67	67 m/sec
Flow Rate	400	400 gpm
SPP	1194	1908 psi
% Pressure @ Bit	50.4%	31.5%
Measured Depth	1378	3707 m
TVD	1378	3707 m
Mud Weight	14.17	14.17 ppg
Plastic Viscosity	10	15 cp
Yield Point	15	27 lbs/100 ft²

Drill String And Annular Velocity Data @ 3707 m

Depth m	OD X ID in	Length m	Hole Size in	Bore Loss psi	Ann. Loss psi	Ann. Vel. m/min	Est. Chip Vel. m/min	Crit. Vel. m/min	Flow Type
0-3525	DP 5 x 4.24	3525	8.500	635.1	307.8	63	53	89	LAM
3525-3552	HWDP 5 x 3	27	8.500	25.8	2.4	63	53	89	LAM
3552-3553	6.5 x 3	1	8.500	1.0	.2	100	89	93	TRANS
3553-3580	DC 6.5 x 2.813	27	8.500	35.2	6.4	100	89	93	TRANS
3580-3588	6.5 x 2.75	8	8.500	11.6	1.9	100	89	93	TRANS
3588-3707	DC 6.5 x 2.813	119	8.500	154.9	28.4	100	89	93	TRANS

6.3 5-7/8" BHA

BIT SIZE (inch)	5-7/8"	**BHA Configuration to C/Out Inside 7" Liner**				**Collar**	**OD**
Depth In/Out (m)	4663					DC-1	4-3/4"
HOLE INCLINATION	0						
MUD WEIGHT (pcf)	106						
B.FACTOR	0/784						
Weight in Air (lb)	29,642						
Bouyed Weight (lb)	23,229						
Bouyed Weight Below Jar (lb)	16,462						

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4/276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	344/61
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115/2	343/61
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54/84	228/41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27/42	173/57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	146/15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137/1	138/15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0/75	1/05
	5-7/8" MT Bit	1	5-7/8	-	3-1/2" Reg. P x B	0/3	0/3

7. Mud & Cement Program

7.1 Mud Program

Mud Properties				AZADEGAN
Fluid Density (pcf)	106		Fluid System	Salt saturated mud
Hole Size (inch)	Completion Phase			
Casing Depth (m)	3854		Solids Control Equip Eff.%	
Casing OD/ ID (inch)	9 5/8	8.681	Well TD (m)	5340
Liner Depth (m)	4663		Liner Lap (m)	3707.5
Liner OD / ID (inch)	7	6.094	Liner Length (m)	955.5

Volumes Information	Qty (bbl)
Surface Volume [Considering dead volumes of pits and lines while transferring]	400
Cased Hole Volume [Capacity of 3707.5m 9⅝" Casing + Capacity of 955.5m 7" Liner]	1004
Dilution	500
Total New Volume Required (bbl)	**1904**

Estimated Chemical Consumption and Cost based on 106 pcf MW					
PRODUCT	**UNIT SIZE**	**CONC'n**	**Expected Consump. for Mixing**	**Expected Consump. for Treating**	**Total Qty Expected To Be Used**
SEA WATER	1 bbl	0.746	1,420		1420
CAUSTIC SODA	25 KG	0.3	10	5	15
SODA ASH	50 KG	0.3	5	5	10
HYDRO-ZAN	25 KG	0.5	17	24	41
BARITE	1 MT	245	212	106	318
Materials Required for Probable Contingency Situations					

Pipe-Freeing Agent	55 Gal DM	Contingency	8	8
Lime	25 Kg Sx	Contingency	80	80
Zinc Carbonate	25 Kg Sx	Contingency	120	120
Sodium Bicarbonate	25 Kg Sx	Contingency	360	360
Citric Acid	25 Kg Sx	Contingency	200	200
Defoamer A	55 Gal DM	Contingency	4	4
Bio-Cide	55 Gal DM	Cement Drill Out	4	4
Drilling Detergent	55 Gal DM	Contingency	4	4
Corrosion Inhibitor	55 Gal DM	Contingency	4	4

NOTE:

- Above calculation are based on normal hole condition
- Above Calculated is Based on **106 pcf**

Completion Phase (Fluids#1&2)			
Fluid Density (pcf)	105	**Fluid System**	**COMPLETION & PACKER FLUID $CaBr_2$ Brine**
Hole Size (inch)	Completion Phase		
Casing Depth (m)	3854	**Solids Control Equip Eff.%**	
Casing OD/ ID (inch)	9-5/8 / 8.681	**Well TD (m)**	5340
Liner Depth (m)	4392	**Liner Lap (m)**	3707.5
Liner OD / ID (inch)	7 / 6.094	**Liner Length (m)**	684.5

Volumes Information	**Qty (bbl)**
Surface Volume [Considering dead volumes of pits and lines while transferring]	250
Cased Hole Volume [Capacity of 3707.5m 9⅝" Casing + Capacity of 684.5m 7" Liner]	970
Total New Volume Required (bbl)	**1220**

Estimated Chemical Consumption and Cost based on 105 pcf MW							
PRODUCT	**UNIT SIZE**	**UNIT PRICE (EURO)**	**CONC'n**	**Expected Consump. for Mixing**	**Expected Consump. for Treating**	**Total Qty Expected To Be Used**	**TOTAL COST (EURO)**
Calcium Bromide (14.2 ppg)	55 Gal	N/A	0.928 bbl/bbl	892	66	958	N/A
Corrosion Inhibitor	55 Gal	N/A	1% Vol	3	1	10	N/A
H2S Scavenger Liquid	55 Gal	N/A	0.5% Vol	5	3	8	N/A
Oxygen Scavenger CA	25 Kg	N/A	0.5 ppb	9	6	16	N/A

Table 9: Mud Properties

NOTE:

- Above calculation are based on normal hole condition
- Above Calculated is based on **105 pcf Mw.**

Completion Phase (Fluid#3)					
Fluid Density (pcf)	86			**Fluid System**	**HI-VIS PILL CaCl2 Brine 86 pcf + HEC**
Hole Size (inch)	Completion Phase				
Casing Depth (m)	3854			**Solids Control Equip Eff.%**	
Casing OD/ ID (inch)	9-5/8	/	8.535	**Well TD (m)**	5340
Liner Depth (m)	4392			**Liner Lap (m)**	3707.5
Liner OD / ID (inch)	7	/	6.094	**Liner Length (m)**	684.5

Volumes Information	Qty (bbl)
Hi-Vis Brine	**150**
Total New Volume Required (bbl)	**150**

Estimated Chemical Consumption and Cost based on 86 pcf MW

PRODUCT	UNIT SIZE	UNIT PRICE (EURO)	CONC'n	Expected Consump. for Mixing	Expected Consump. for Treating	Total Qty Expected To Be Used	TOTAL COST (EURO)
Fresh Water	55 Gal	N/A	0.823 bbl/bbl	124		124	N/A
CaCl2 95%	25 Kg	N/A	195 ppb	531	80	611	N/A
HEC	25 Kg	N/A	1 ppb	3	3	6	N/A

NOTE:

- Above calculation are based on normal hole condition.
- Above Calculated is based on **86 pcf Mw.**

Completion Phase (Fluids#4)

Fluid Density (pcf)	86		**Fluid System**	**SPACER CaCl2 Brine 86 pcf**
Hole Size (inch)	Completion Phase			
Casing Depth (m)	3854		**Solids Control Equip Eff.%**	
Casing OD/ ID (inch)	9-5/8	8.535	**Well TD (m)**	5340
Liner Depth (m)	4392		**Liner Lap (m)**	3707.5
Liner OD / ID (inch)	7	6.094	**Liner Length (m)**	684.5

Volumes Information	Qty (bbl)
CaCl2 Brine 86 pcf	**150**
Total New Volume Required (bbl)	**150**

Estimated Chemical Consumption and Cost based on 86 pcf MW

PRODUCT	UNIT SIZE	UNIT PRICE (EURO)	CONC'n	Expected Consump. for Mixing	Expected Consump. for Treating	Total Qty Expected To Be Used	TOTAL COST (EURO)
Fresh Water	55 Gal	N/A	0.823 bbl/bbl	124		124	N/A
CaCl2 95%	25 Kg	N/A	195 ppb	531	80	611	N/A

NOTE:

- Above calculation are based on normal hole condition.
- Above Calculated is based on **86 pcf Mw.**

7.2 Cement Plug Program

Cement program will be issued in due time.

8. Perforation Program

Displace 7" Liner volume with CaBr2 perforation fluid, perform 7" Liner perforation, run CCL/GR log to confirm perforated interval. RIH with 5 7/8" Bit to top of perforation while scrap 7" production PKR setting depth, displace well volume with CaBr2 completion fluid. Run 4-1/2" * 3-1/2", 10K Completion string. N/D BOP, N/U XMT. Perform well Pre-Acid clean up, Acid-Wash, Post-Acid clean up and well performance test operations.

The perforation intervals will be in Fahliyan Formation Fah-1a, Fah-1b and Fah-1c with a thickness of 14.7m, 15.7m and 32.4m TVD respectively. The Contractor must prepare the detail of perforation program under supervision of client.

8.1 Perforation Gun

Wireline conveyed perforation can be the first priority, and TCP (tubing conveyed perforating) to be considered as back-up plan in case of tough conditions in the wellbore.

8.2 Perforation Gun and Charges/CCL-GR

Perforation gun and charges shall be provided according to Table 7. CCL-GR shall be provided in requested sizes. Guns can be run on wireline or TCP be performed depending on total perforation length.

Table 10: proposed perforation Gun

Gun OD (in)	Shot density (SPF)	Phase angle	API Penetration Depth (in)	Explosive type	Under-balance pressure(psi)
4.5	5	72°	≥40	HMX	NA

9. Completion Program

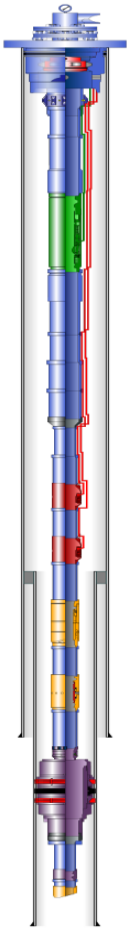
9.1 Existing Completion string

There is no completion string in well. The well was suspended by BP & cement plugs.

9.2 Propose Completion string

AZNS-010 Completion Sketch

Contractor		Liner & Tubing Data					Client
Field	AZADEGAN	Description	7"Liner	3 1/2" Tubing	4 1/2" Tubing	Tubing	7" Liner Lap: 3707.5
Well Number:	AZNS-010	OD[in]	7/00	3/50	4/50		7" Liner Shoe: 4922
Rig :	201	Nominal weight [lb/ft]	32/00	10/20	18/90		5' Liner Lap: -
RTE-GLE	TBA	Grade	p-110	L-80	L-80		5" Liner Shoe -
RTE-THF	TBA	Thread	PJE	Premium	Premium		Plug Back Depth 4392
Rig Spudded Date :	TBA	Inside diameter [in]	6/184	2/92	3/640		X-mass Tree: -
End of Operation	TBA	API Drift Diameter	6/059	2/80	3/515		Size & Rating:

Schematic		Description	Max OD	Min ID	Length(m)	BTM(m)	TOP(m)
	1	Elevation (floor - TBG Head Flange)			11/50		
	2	TBG HANGER 11" * 4 1/2" ACME Box up * 4 1/2" Premium Box Down , M/C: FF-0.5 T/C: U PSL: 3G PR: 2, 4" Type "H" Two Way	11	4	0/65	0/650	0/000
	3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# Premium* SUPER 13CR-110KSI	6/1	3/64	1/83	2/480	0/650
	4	5 JOINTs TUBING 4 1/2", Premium ,18.9#, L-80, Type-1	5/118	3/64	55/00	57/480	2/480
	5	PUP JOINT 4 1/2", Premium ,18.9#, L-80, Type-1	5/118	3/64	2/00	59/480	57/480
	6	4-1/2" FLOW COUPLING,18.9#,SUPER 13CR-110KSI,Premium , 6 FT	5/25	3/64	0/82	60/300	59/480
	7	TRSSV-10 Tubing Retrievable Saftey Valve, 4 1/2", 18.9# W/ 3.437" Seal bore, Premium , SUPER 13CR-110KSI, 10K WP With AF Profile, Flapper Type,	7/5	3/437	1/85	62/150	60/300
	8	4-1/2" FLOW COUPLING, 18.9#,SUPER 13CR-110KSI, Vam Top , 4' FT	5/25	3/64	1/21	63/360	62/150
	9	320 JOINTs TUBING 4-1/2"Premium 18.9#, L-80	5/118	3/64	3520/00	3583/360	63/360
	10	X-over 4-1/2"Premium, 18.9# BOX * 3-1/2"Premium , 10.2# PIN,	5/25	2/922	0/61	3583/970	3583/360
	11	14 JOINTs TUBING 3-1/2"Premium , 10.2#, L-80.Type-1	4/25	2/922	150/00	3733/970	3583/970
	12	3-1/2" PUP JOINT, Premium , 10.2#, L-80, Type-1	4/25	2/922	2/00	3735/970	3733/970
	13	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDREL Premium 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2/922	0/90	3736/870	3735/970
	14	3-1/2" PUP JOINT, Premium , 10.2#, L-80, Type-1	4/25	2/922	2/00	3738/870	3736/870
	15	1 JOINT TUBING 3-1/2"Premium , 10.2#, L-80.Type-1	4/25	2/922	11/00	3749/870	3738/870
	16	3-1/2" PUP JOINT, Premium , 10.2#, L-80, Type-1	4/25	2/922	2/00	3751/870	3749/870
	17	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELPremium , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2/922	0/90	3752/770	3751/870
	18	3-1/2" PUP JOINT, Premium , 10.2#, L-80, Type-1	4/25	2/922	2/00	3754/770	3752/770
	19	1 JOINT TUBING 3-1/2" Premium , 10.2#, L-80.Type-1	4/25	2/922	11/00	3765/770	3754/770
	20	3-1/2" EXP. JOINT, Premium , SUPER 13CR-110KSI, 10'STK (Half Closed) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS,	4/906	2/922	6/50	3772/270	3765/770
	21	1 JOINT TUBING 3-1/2" Premium , 10.2#, L-80.Type-1	4/250	2/922	11/00	3783/270	3772/270
	22	3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-110KSI, With Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN)	4/551	2/75	1/45	3784/720	3783/270
	23	1 JOINT TUBING 3-1/2" Premium , 10.2#, L-80.Type-1	4/250	2/922	11/00	3795/720	3784/720
	24	3-1/2" ANCHOR SEAL ASSEMBLY, Premium, 10.2#,SUPER 13CR-110KSI, 4" PACKING OD		2/922	0/74	3796/460	3795/720
	25	7" HYDRAULIC SET PERMANENT PRODUCTION PACKER, MODEL WSDP TO BE RUN IN 7.00" 29-32 PPF CASING, SUPER 13CR-110KSI, UPPER SEAL BORE: 4"	5/750	2/922	1/70	3798/160	3796/460
	26	4-1/2" MILL OUT EXTENSION, 18.9#,Premium , SUPER 13CR-110KSI	4/500	3/64	1/70	3799/860	3798/160
	27	X-over 4-1/2" Premium 18.9# BOX * 3-1/2" Premium , 10.2# PIN, SUPER 13CR-110KSI	5/250	2/922	0/60	3800/460	3799/860
	28	1 JOINT TUBING 3-1/2" Premium , 10.2#, L-80.Type-1	4/25	2/922	11/00	3811/460	3800/460
	29	3-1/2" BOTTOM NO-GO LANDING, Premium , 10.2#, SUPER 13CR-110KSI W/ Seal Bore 2.562" TYPE XN (NO-GO ID: 2.442")	4/25	2/562	0/50	3811/960	3811/460
	30	3-1/2" WIRLINE RE-ENTRY GUIDE, 10.2 #, Premium Con,SUPER 13CR-110KSI	4/25	2/922	0/45	3812/410	3811/960
		END OF COMPLETION STRING					3812/410

3SV Landing Nipple	62/150	60/300
Traveling Joint	3772/270	3765/770
CIV 1	3736/870	3735/970
CIV 2	3752/770	3751/870
SSD	3784/720	3783/270
7" Retainer Packer	3798/160	3796/460
NO-GO Landing Nipple	3811/960	3811/460
Entry Guide	3812/410	3811/960

Figure 7: Propose Completion Schematic

10. BOP & X-Mass Tree

10.1 Existing X-Mass Tree

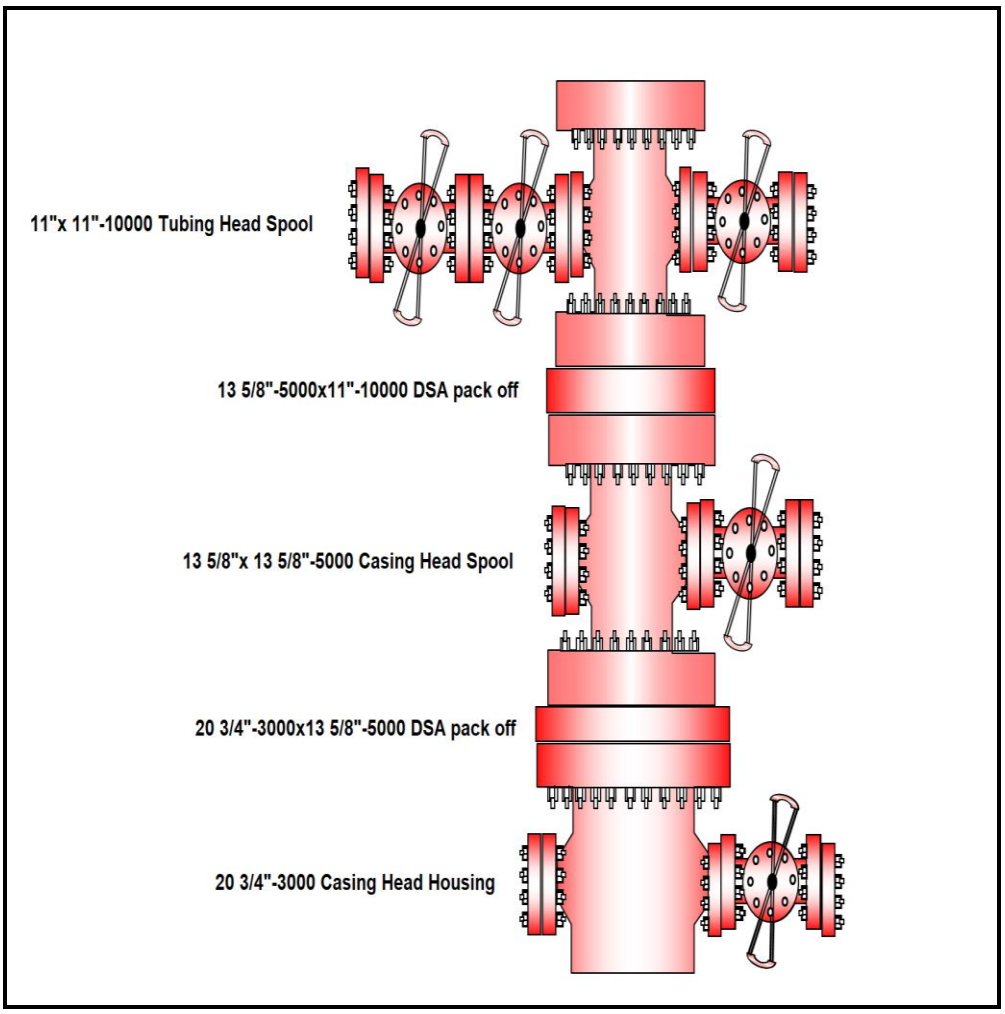


Figure 8: Existing Wellhead

10.2 Propose X-Mass Tree

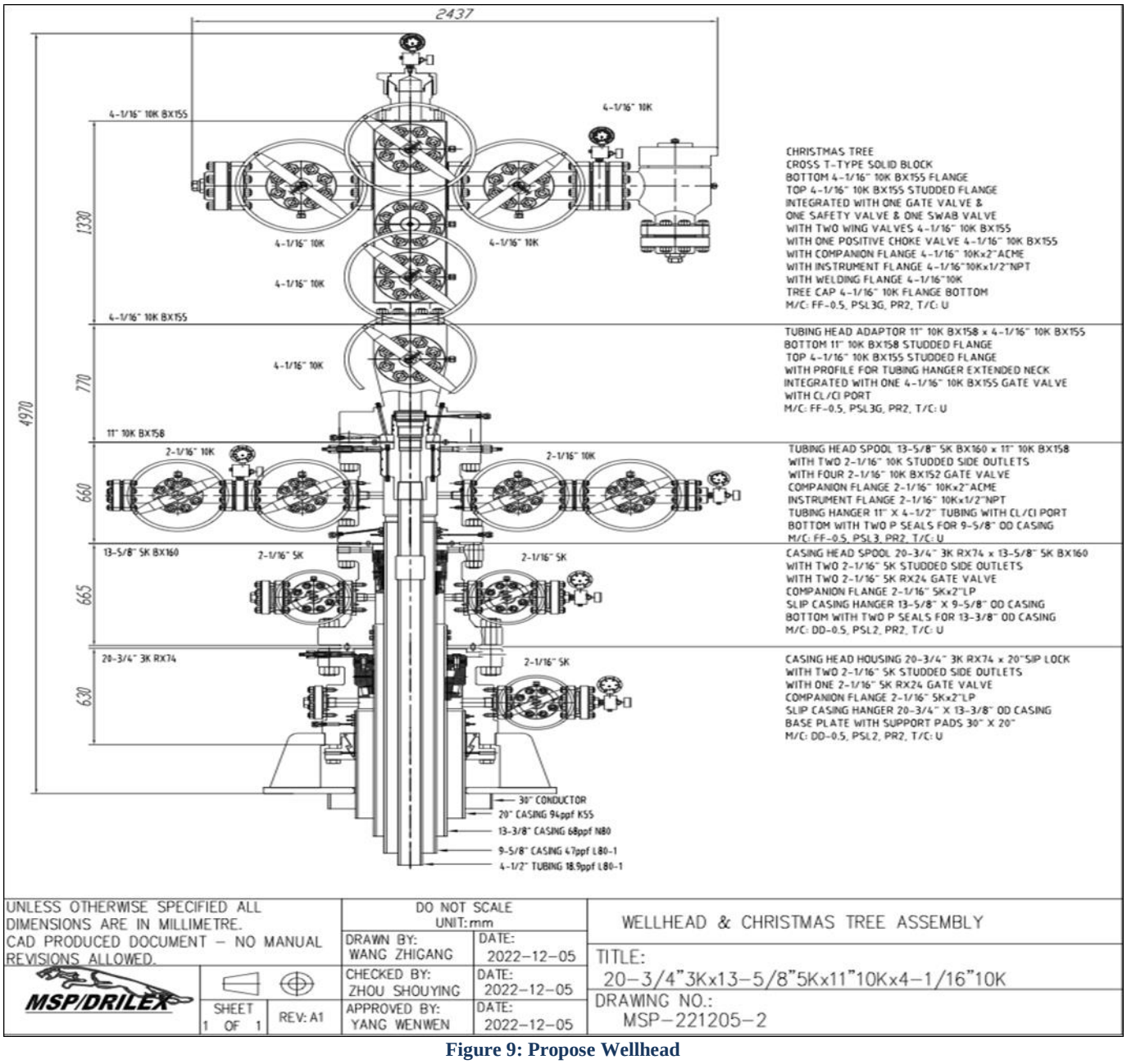


Figure 9: Propose Wellhead

Item Well type	Casing Head Housing	Casing Spool	Tubing Spool	X-mas Tree
Savark	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5 K	5K
Gadvan / Fahliyan	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 10K × 13 5/8" 5 K	10K
Water disposal wells	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5K	5K

Item	Specification
Casing Head Housing assembly	MC:DD-0.5 3000PSI PSL-2 PR-2, TC: U
Casing Spool assembly	MC:DD-0.5 5000PSI PSL-2 PR2, TC: U
Tubing Spool assembly	MC: FF-05 5000PSI PSL-3, PR-2, TC: U (Savark & Water disposal wells)
	MC: FF-0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)
Tubing hanger	MC: FF 0.5 disposal wells 5000PSI PSL-3G, PR2, TC: U (Savark & water)
	MC: FF 0.5 10000PSI PSL-3G, PR2, TC: U (K&G&F wells)
X-Tree assembly	MC: FF 0.5 5000PSI PSL-3G, PR-2, TC: U (Savark & Water disposal wells)
	MC: FF 0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)

Table 11: AZNS Wellhead Information

- **Wellhead Specification**

NOTE: Section “A” and “B” already installed by Client, section C will be changed and section D will be installed by contractor.

20-3/4" x 13-5/8" x 11" x 4-1/16" x 10,000 psi WP for Fahliyan wells	
Section A	**Casing head housing** for 20” (94lb/ft) OD casing x 20 3/4” 3,000 psi API flange **has hold down bolts for wear bushing,** with 2 x 2- 1/16” 3000 psi studded side outlets, test ports and seals. One side outlet has **one** 2 1/16”x 3,000 psi manual gate valve, 2” LP threaded companion flange and 2” bull plug threaded for ½” NPT and fitted with 0-3,000 psi gauge and needle valve and, the other side to have 2 1/16” x 3,000 psi x 2” LP threaded flange and 2” bull plug. Casing slip and seal assembly self-energizing to hang 13 3/8” (68lb/ft) OD casing. This section is installed on 20” casing stub by way of **slip lock** with landing plate on 30" conductor pipe. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 630mm** Material: Forged alloy steel 4130 Automatic casing slip material: Low alloy steel 4130
Section B	**Casing head spool** with 13 5/8” 5,000 psi top flange x 20 3/4” 3,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 13 3/8” OD casing with test ports to test annular space and casing seals. One side outlet has **one** 2 1/16”x 5,000psi manual gate valve, 2” LP threaded companion flange and 2” bull plug on one side and, the other side has **one** 2 1/16”x 5,000psi manual gate valve, 2” LP threaded companion flange, 2” bull plug threaded for ½” NPT and fitted with 0-5,000psi gauge and needle valve. Casing slip and seal assembly self-energizing to hang 9 5/8” (47lb/ft) OD casing. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 665mm** Material: Forged low alloy steel 4130 13 5/8”x 9 5/8” automatic casing slip material: Low alloy steel 4130
Section C	**Tubing head spool** with 11” 10,000 psi top flange x 13 5/8” 5,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 9 5/8” OD casing with test ports to test annular space and casing seals C/W T.D.Screws with 410SS material Side outlets have 1.900" sharp vee VR threads on both sides, to have **two** 2 1/16”x 10,000psi manual gate valve, 2”ACME 10K female thread companion flange, 2” bull plug threaded for ½”NPT and fitted with 0-10,000psi gauge and needle valve on one side and, the other side to have the first 2 1/16”x 10,000psi manual gate valve, 2 1/16" 10,000 psi instrument flange installed with 0ne ½”NPT and fitted with 0-10,000psi gauge and needle valve, and the second 2 1/16" x 10,000 psi manual gate valve, 2” ACME 10K female thread companion flange and 2” bull plug.(manual G.valve to be non-rising stem w/back seat capability). Clear bore for passing 8-1/2" BHA. The 11" x 4 1/2" **tubing hanger** has 4 1/2" 18.9ppf premium conn. Top & bottom thread one continuous 1/4" NPT control line exit for SSSV, two channels for **3/8"** chemical injection line exit, one 1/4" line exit for DHPG. 4" type "H" BPV thread preparation, one metal seal & two S-SEALS to pack off the upper neck and two S-SEAL to pack off the body of hanger, and tonging preparation on the lower neck. Tubing hanger shall have factory-torqued double pin tubing pup joint 4 1/2" 18.9 ppf, L80, New Vam approximate length 1600mm to facilitate makeup. **Material FF-0.5, PSL-3, PR-2 with exception to tubing hanger: FF 0.5, PSL-3G, PR-2, Temp U. Height: 850mm**

	Material of body: Forged low alloy steel 410SS G. valves, material of seats/gates & stem: Forged alloy steel AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.
Section D	**Tubing head adaptor (w/one integral manual G. valve)** 11" 10,000psi studded flange x 4 1/16" 10,000psi full bore manual gate valve with bottom preparation to seal on tubing hanger (one metal seal & 2 S. seal) and with test ports to test annular space and seals, top prepared with one continuous 1/4" NPT control line exit, one 1/4" line exit for DHPG and two channels for **3/8"** chemical injection line exit. (Manual G. valve to be non-rising stem w/back seat capability) The T-TYPE **Christmas tree** consists of the following: (1) **Solid block valve (T-Type):** 4 1/16" 10000psi bottom flange for tubing adaptor. 4 1/16" 10000psi lower manual gate valve, 4 1/16" 10000psi upper gate valve with SSV w/ hydraulic actuator, including wire line cutting function & lock open cap, (2) 4 1/16" 10000psi manual swab valve. With two 4 1/16" 10000psi studded outside for wing valves. With one 4 1/16" 10000psi studded top flange for tree cap. (3) one wing valve 4 1/16" 10,000 psi full-bore manual gate valve, one **instrument flange** with one 1/2" NPT ports c/w one 0-10,000 psi pressure gauge and needle valve installed and one 4 1/16"10,000psi Cage type positive choke with 90 deg outlet.(With 2 no of each Wrench): positive choke with 2 sets of bean sizes 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 128/64 inch and with welding flange prepared for 4" pipe(specification to be confirmed). (4) The other wing with one 4 1/16" 10,000psi full-bore manual gate valve and one companion flange w/2" ACME 10K female thread. (5) Quick union type Tree Cap above, with 4 1/16" 10,000psi flange at bottom, ACME thread union on top, including 1/2"NPT port for needle valve and pressure gauge. **Material FF-0.5, PSL-3G, PR-2, Temp U. Height of integral tubing head adaptor and gate valve=800mm**. Material of body: 410ss G. valves, material of seats/gates & stem: Forged alloy steel, AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

Table 12: AZN 10 Wellhead Specification

10.3 BOP

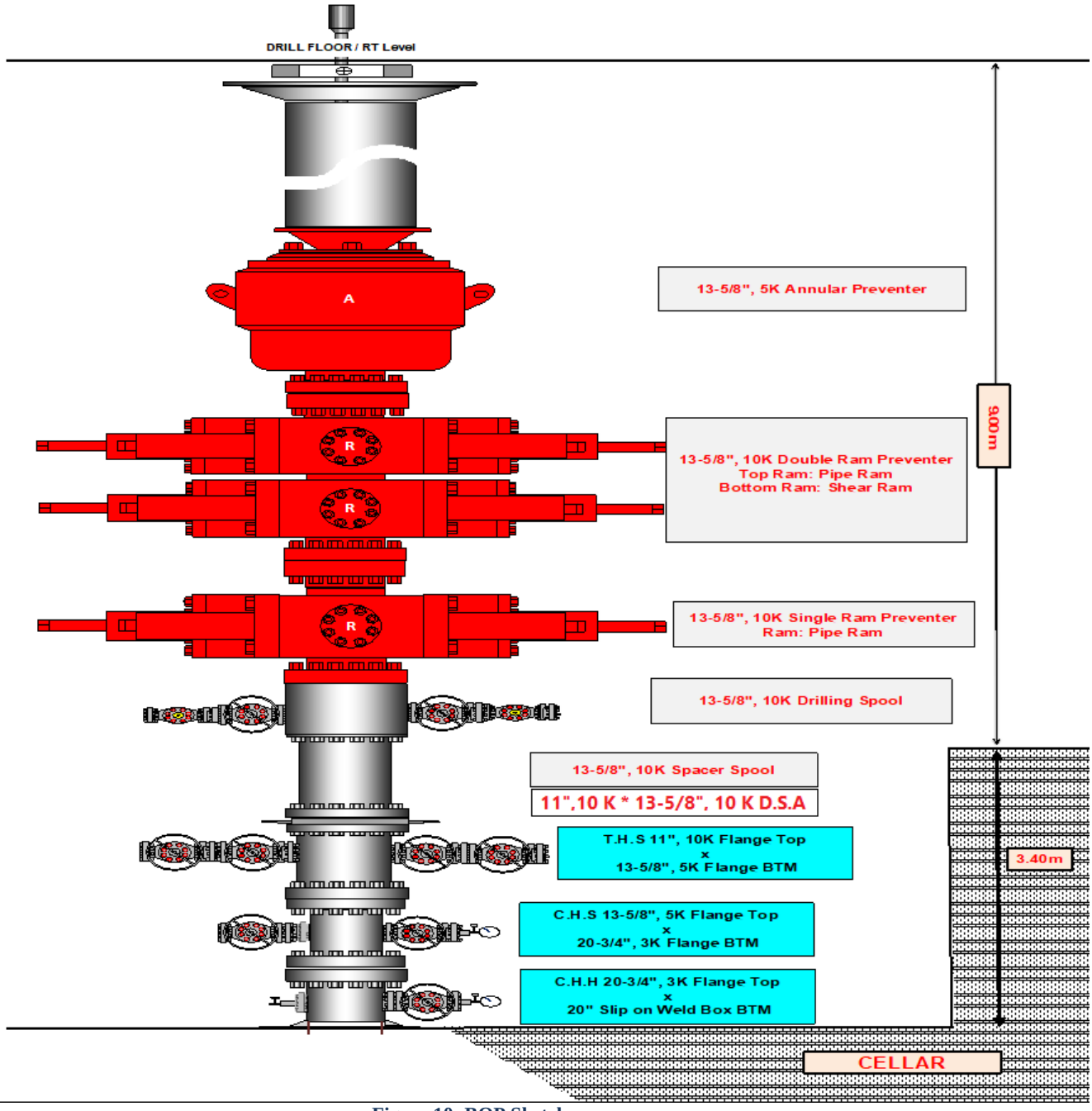


Figure 10: BOP Sketch

11.

11. Waste Management

11.1 Scope of work

11.1.1 Drilling waste management design

The contractor is responsible for designing the optimum practices for treating the produced waste during the operation. The initial technical waste management proposal should be included the practices for minimizing, treating and recovering of the wastes.
According to the below categories:

- Water based mud containing minerals and natural synthetic polymers, and etc.
- Cuttings contaminated with WBM.
- Oil Based Mud/ Oil Emulsion Mud.
- Cuttings contaminated with OBM.

The contractor is encouraged to offer possible practices to minimize the drilling fluids waste and its regeneration.

11.1.1.1 WATER BASED MUD CONTAINING MINERALS, NATURALS AND SYNTHETIC POLYMERS

- **Primary operation**

The proposed primary solids control package should be able to control drilling fluid properties, reducing of mud materials usage against conventional solid control equipment's, and significantly reduce the number of excess fluids generated throughout drilling each phase of the well.

- **Secondary Operation**

All excess mud and spillage (water based) should be collected in rig site pits and free surface fluid should be treated using a built for propose flocculation station (has to join with centrifuge) consisting of integral dosing and pumps along with a centrifuge for efficient, quick and controlled separation of the solids before the cleaned fluids is returned to active system or buffer tanks.
This system should be able to remove colloidal sized particles that cannot be removed with primary solids control system.
This system may use to clean up the remaining water-based mud or decrease weight of oil base mud at the end of the drilling fluids programmed.

11.1.1.2 CUTTINGS CONTAMINATED WITH WBM

Spud mud and water-based mud cuttings should be fixed and be ready for transport to disposal zone. For fixing the cutting, 100-150 kg cement shall be used for each cubic meters of cutting as well as 3-4-liter Sodium Silicate for each cubic meter of cutting. (Consumption of cement and sodium silicate varies in different formations). Gypsum is not allowed for cutting fixation. This method is physically Bonds Contaminants to Prevent Leaching to Environment. Final product will pass can test and in case of approval by company man it'll send to landfill.

11.1.1.3 OIL BASED MUD/ OIL EMULSION MUD

Oil based fluids after treated in solid control and removal solids stored for future re-use as drilling fluids by responsibility of company (PPL).

The density of oil base fluids for re-used or storage should be adjusted according to client requested weight.

11.1.1.4 CUTTINGS CONTAMINATED WITH OBM

Cuttings that are not eligible for spreading should be stabilized with OSEII method to ensure a stable mix, then buried and dispose according to environmental policy.

- **BPD I Method**

BIOLOGICAL PARD DRILLERS I is a unique Bio catalytic System of preformed multi-enzyme liquid concentrate. BPD stimulates and accelerates natural biological reactions. When combined with fresh or salt water and oxygen, BPD will cause crude oil and other organic substances to rapidly decompose; eventually biodegrading them to carbon dioxide and water.

BPD is non-toxic to humans, animals, plants and marine life. It is nonpoisonous, even if accidentally ingested. It is non-irritating to the most sensitive skin. BPD contains no known allergens to cause skin, respiratory or other allergic reactions. Birds bathed in BPD should be quarantined until their own natural oils are restored.

Once BPD is applied to a hydrocarbon spill, the enzymes and other product constituents start emulsification and solubilization of the hydrocarbon substrate. Emulsification and solubilization generally take from a few minutes up to a few hours for weathered heavy-end hydrocarbons, once BPD is applied with a temperature of 40 degrees F. or greater. Once solubilization is completed, the hydrocarbon substrate is less toxic (and the hazard of fire is diminished) so the enhanced - naturally occurring bacteria will have a higher affinity for the solubilized hydrocarbon substrate.

11.1.2 Waste discharge and handling

The Company is responsible for designing of required pits and The Contractor is responsible for procedure for burring of waste cuttings in disposal zone; supervision and managing of requirement operation for deposit oil waste cutting.

The contractor is responsible for treating all the generated waste in site without any discharges in environment or damages. It is also the responsibility of the contractor for handling/discharge of generated waste after treatment.

The contractor manages and supervises the solids and fluid waste streams loading and off-loading on/from the designated areas and responsible for the loader along with the responsibility of the storage pits, tanks, etc.

Note: All mentioned operation that will be done by contractor as extra work.

11.1.3 Laboratory equipment's and analysis

The contractor should prepare suitable laboratory analysis equipment's to perform complete analysis services of waste matter. The provided equipment's shall make this opportunity to study the mentioned environmental policy of waste analysis.

11.1.4 Reports and comments

The contractor provides the following reports and documentations on contract duration:

- Daly and monthly waste management report.
- Final waste management report, including:
  - Summarized amount of generated waste on each section,
  - Total treated waste,
  - Total chemical consumption,
  - Cost monitoring report, and
  - Special activities.

11.1.5 The equipment's installation and commission

- First location

The Maximum days for installation and commissioning of all equipment and necessary piping will be 7 days after rig spud. Contractor is responsible for any delay to installation and commissioning of equipment.

- Other locations

The Installation of all equipment should be completed during rig spud. Contractor is responsible for any delay to installation and commissioning of equipment.

11.1.6 Clean up process

After drilling operations and drilling rig removal, the well area, waste pit, and corral should be completely cleaned. The contractor shall remove the fixed cuttings to pre-determine locations (disposal) which can protect the environment is very effective.

11.1.7 Supplying services

Table 13 _ Supplying services.

Waste treatment package . Dewatering unit . Filter press
Transfer pump package . Air diaphragm pump . Submersible pump . Centrifugal pump
Storage tank . Waste water storage tank . Waste water agitators
Cutting treatment . Loader
Waste transportation . Cutting transport truck . Tanker truck for waste water transfer
Other . Wash gun . Laboratory and cabin

11.1.8 Waste management policy

Waste management services should be utilized to minimize generated waste and contaminated discharged cuttings during drilling operations. This helps keep the environment clean and ensures that our operations are in compliance with DOE (Department of Environment) legislations.

We provide cost-effective and optimum solutions from environmentally acceptable procedures to achieve zero-discharge plans and improve fluids recovery. The outcome reduces operational costs and enables recycling.

The waste is analysed to study various parameters such as chloride, nitrate, and phosphate. Other analysis is carried out to monitor treatment processes on a regular basis during waste generation. Our waste management personnel are fully trained experienced and have the competencies in place to provide the best reliable services with concentrated support and the least liabilities.

11.1.9 Drilling waste management steps

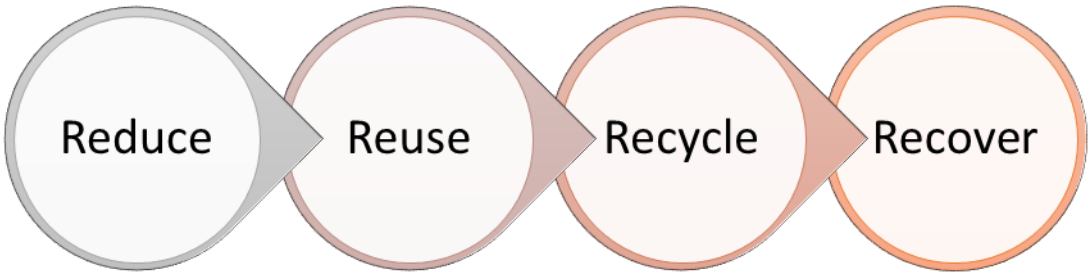


Figure 11 _ Drilling waste management procedure.

11.2 Calculation

11.2.1 Waste water treatment and disposal

11.2.1.1 DEWATERING PROCEDURE

1-The wastewater in the coral (after settling the liquid) is transferred to the buffer tanks.

2-The act of mixing and reacting polymer materials with drilling waste is done in the dewatering unit, which includes a chamber for mixing and mixers, monopumps and reciprocating pumps.

3-The treated wastewater is transferred to the separation unit and its solids are removed by the filter press.

4-Dewatered water is sent to the drilling rig and used again.

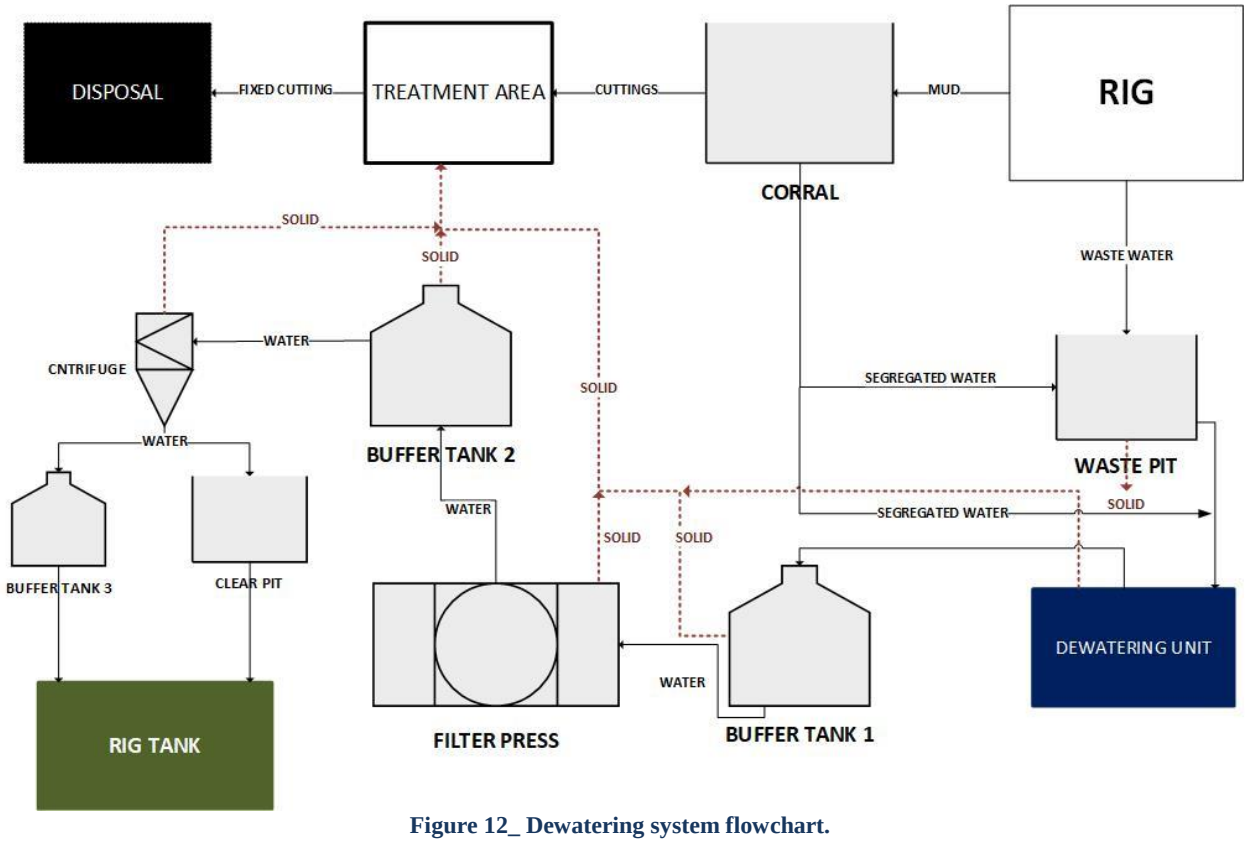


Figure 12_ Dewatering system flowchart.

11.2.1.2 ESTIMATED VOLUMES

- Assumed volumes from drilling mud plan
- The produced daily waste water of 50 m3 and 60% mud are roughly estimated.

Table 14 _ Estimated treating waste water.

	Unit					Total
Hole Size	Inch	**26**	**17.5**	**12.25**	**8.5**	-
Casing Size	Inch	18 5/8	13 3/8	9 5/8	7	-
Depth	Meter	151	1502	3854	5340	-
Mud system	-	Salt saturated mud				-
Mud weight	PCF	106				-
Interval	Meter	151	1351	2352	1486	-
Operating duration	Day(s)	48				**48**
Estimated mud volume	M^3	372				**372**
	bbl.	2304				**2304**
Estimated Treating waste water	M^3	2546				**2546**
	bbl.	15782				**15782**

11.2.1.3 DEWATERING CHEMICALS

Table 15_ Dewatering chemical.

	Application	Required concentration
Poly Aluminium chloride	Coagulant	0.5~1.5 Kg/m^3
Flocculent polymer	Flocculent	0.1~0.3 Kg/m^3
Hydrated lime	Coagulant	0.25~0.5 Kg/m^3
Soda-ash	Harness control	0.5 ~ 0.6 Kg/m^3 55increasing 1000 mg/Lit of Ca, at average
Caustic soda	pH control	0.5 ~ 0.6 Kg/m^3 test depended
Biocide	Microbial activity control	20 ml/ m^3

11.2.1.4 TREATED WATER PROPERTIES

Table 16 _ Treated water properties.

Item	Amount
pH	7~9
Total hardness	< 800 mg/liter
Turbidity	< 50 NTU
Nitrate	< 50 mg/liter
Aluminium	< 3 mg/Lit
Sulphate	< 50 mg/liter
Ca^{2+}	< 600 mg/liter

11.2.1.5 MATERIAL CONSUMPTION ESTIMATION

This estimation, does not eliminate the responsibility of the test operator.

Table 17 _ Material consumption.

	Unit					Total
Hole Size	Inch	**26**	**17.5**	**12.25**	**8.5**	-
Casing Size	Inch	18 5/8	13 3/8	9 5/8	7	-
Depth	Meter	151	1502	3854	5340	-
Interval	Meter	151	1351	2352	1486	-
Operating duration	Day(s)			48		**48**
Poly alum	Dosage of kg/M^3			0.5~1.5		-
	25 kg sxs			101		**101**
Poly electrolyte	Dosage of kg/M^3			0.1~0.3		
	25 kg sxs			20.36		**20.36**
Soda ash	Dosage of kg/M^3			1		
	25 kg sxs			101		**101**
Caustic soda	Dosage of kg/M^3			0.5~0.6		
	50 kg sxs			76.36		**76.36**
Biocide	Dosage of ml/M^3			20		
	55Gallon/Dr			2.5		**2.5**
Lime	Dosage of kg/M^3			0.5~1		
	25 kg sxs			101		**101**
Citric acid	Dosage of kg/M^3			0.3~0.5		
	25 kg sxs			30		**30**

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11.2.2 Cutting treatment

11.2.2.1 CUTTING REMOVAL PROCESS

1-Salt saturated mud displaced by oil-based mud and crude oil and discharge oily waste.
2-Oil base mud and crude oil are separated by solids control devices (shale shaker, desander and desilter) and this work is done by the company1.
3-Solid waste includes OBM and crude oil are discharged into the corral
4-OBM fluids and crude oil are used (after separation) with the company's responsibility for industrial purposes or reused for oil base mud
5-Solid waste includes OBM and crude oil are treated and disposed by biological methods and after lower tph, fixed them (BPD I)
6-Water based mud discharge in corral, transfer to the treated area and treated by fixation method
7-WBM treated by cement and sodium silicate
8-Transfer WBM, OBM solid waste after treated by mentioned methods to the disposal area which is introduced by the company.

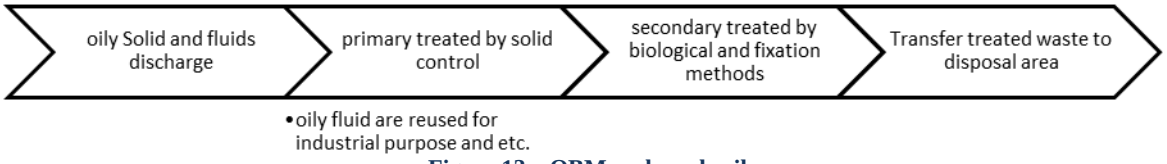


Figure 13 _ OBM and crude oil process



Figure 14 _ WBM fixation process

11.2.2.2 ESTIMATED CUTTING VOLUME

Table 18 _ Estimated cutting volume.

	Unit					Total
Hole Size	Inch	**26**	**17.5**	**12.25**	**8.5**	-
Casing Size	Inch	18 5/8	13 3/8	9 5/8	7	-
Depth	Meter	151	1502	3854	5340	-
Interval	Meter	151	1351	2352	1486	-
Operating duration	Day(s)			48		**48**

[1] Mahan sanat arya

Mud system	-	Salt saturated mud	-
Mud weight	PCF	106	-
Theorical hole cuttings	M^3	Not drilling formation(workover well)	**0**
	bbl.		**0**
Real hole cuttings	M^3	Not drilling formation(workover well)	**0**
	bbl.		**0**
Real mud solid	M^3	149	**149**
	bbl.	922	**922**
Total waste	M^3	149	**149**
	bbl.	922	**922**

This volume estimate based on drilling plan and possible cases should be included in future revisions

11.2.2.3 ESTIMATED FIXATION VOLUME

Table 19_Estimated fixation volume.

	unit					Total
Hole Size	Inch	**26**	**17.5**	**12.25**	**8.5**	-
Casing Size	Inch	18 5/8	13 3/8	9 5/8	7	-
Depth	Meter	151	1502	3854	5340	-
Interval	Meter	151	1351	2352	1486	-
Operating duration	Day(s)		48			**48**
Mud system	-		Salt saturated mud			
Mud weight	PCF		106			
Total waste	M^3		149			**149**
	bbl.		922			**922**
Cement	Kg/M^3		50			**-**
	50 kg		149			**149**
Silica	Kg/M^3		3			**-**
	55Gallon/Dr		2.1			**2.1**

These volumes of chemical estimate only for fixation procedure and every other possible case, including the discharge of oil mud, has not been included in this program

11.3 Waste management equipment's

11.3.1 Flocculation unit

This unit, has the bellow specifications:

- Fully automated system,
- Full 316 stainless steel material,
- Equipped with high pressure level and pH sensors,
- Equipped with digital flow meter and totalizer to monitor flow rate, and
- Touch panel technology to control prose.



Figure 15 _ Flocculation unit.

11.3.2 400bbl Dewatering unit

This tank provides more than 400 bbl. capacity to store contaminated water for using in dewatering process and ability to adding chemicals such lime from hopper. Two Mixer 25 hp and Jet System give ability to holding any type of mud for hours without any stealing.



Figure 16 _ Conditional tank.

11.3.3 400bbl Storage tank

Each tank provides more than 400 bbl. capacity to store clear water for recycling.

11.3.4 High Performance Filter Unit

Filtration plays an important role in the natural treatment of waste water as it percolates through the soil. It is also a major part of most water treatment. Groundwater that has been softened, or treated through iron and manganese oxidation, requires filtration to remove floc created by coagulation or oxidation processes. Since surface water is subject to run-off and does not undergo natural filtration.

it must be filtered to remove particles and impurities. Filtration can be compared to a sieve or micro-strainer that traps suspended material between the grains of filter media. However, since most suspended particles can easily pass through the spaces between grains of the filter media, straining is the least important process in filtration. Filtration primarily depends on a combination of complex physical and chemical mechanisms, the most important being adsorption. Adsorption is the process of particles sticking onto the surface of the individual filter grains or onto the previously deposited materials. Forces that attract and hold particles to the grains are the same as those that work in coagulation and flocculation. In fact, coagulation and flocculation may occur in the filter bed, especially if coagulation and flocculation before filtration was not properly controlled. Incomplete coagulation can cause serious problems in filter operation.



Figure 17_ High Performance Filter Unit.

11.3.5 Centrifugal pump

This pump can be utilized in different situation. 3 x 4 pump can be used for transferring processed water back to rig water tanks for reuse.

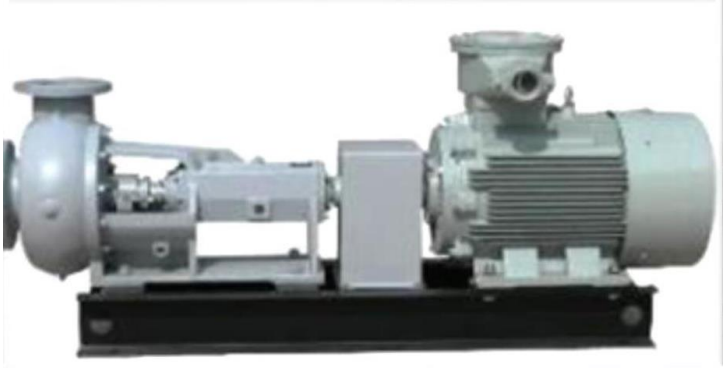


Figure 18 _ Centrifugal pump.

11.3.6 Submersible pump

These pumps designed to clear and waste water from water pits and tanks to designated areas or tanks. They are very easy to handle and can be very useful in case of transferring from bottom of tanks or pits.



Figure 19 _ Centrifugal pump.

11.3.7 200 bars wash down guns

These wash guns minimized water consumption and can be used on the shakers to unplug screens or washing al the area on the rig from rig floor to mud pits.



Figure 20_ 200 bar wash down guns.

11.3.8 Fluid transfer diaphragm pump

This pump works with air pressure and has the responsibility of transferring mud and contaminated water.

Three inches diaphragm pump transfers mud and waste from cellar to coral to keep cellar empty.



Figure 21_ 3” Diaphragm pump.

Two inches diaphragm pumps transfer waste water and mud between different spots on location. They also can be utilizing to transfer mud from mud tanks dead volume to treatment tanks.



Figure 22_ Diaphragm pump

These pumps designed to transfer clear and waste water from water pits and tanks to designated areas or tanks. They are very easy to handle and can be very useful in case of transferring from bottom of tanks or pits.

11.3.9 DWM laboratory cabin

Lab equipment for DWM is consisting of:

- Retort kit,
- API Filtrate test kit,
- Testing reagents and equipment for Chloride, Calcium, Hardness and PH, and
- Hatch Water Test kit





Figure 23_ Hatch Water Test kit

This kit analyses all conserved parameters, including aluminium, iron, nitrate, nitrite, zinc, phosphate, sulfate, sulfide, copper and cyanide concentration.

11.3.10 Storage container



Figure 24 _ Storage container.

11.3.11 Loader



Figure 25 _ Loader.

11.4 Designing pits and treated area

The technical proposal will be presented to the company according to the limitations of the location in the next revisions, and the client is responsible for its implementation.

12. Health, Safety and Environments

12.1 WELL CONTROL

All the drilling operation shall obey the requirement of Well Control Manual of MSA and contractor's well control manual; under no circumstances should the deviation from well control policy be permitted without prior approval through MSA. If the contractor's well control manual or anywhere has conflict with Well Control Manual of MSA, take Well Control Manual of MSA as the standard.

12.1.1 WORK Requirement of Well Control Equipment

- With shut-in casing pressure lower than 1,000psi, while snubbing with annular preventer, drill tools shall have 18-deg shoulder tool joints, and the snubbing velocity shall not be greater than 0.2m/s.
- After shut off the annular preventer or ram preventer, with the shut-in casing pressure not greater than 2,000psi, drill tools are allowed to pull up and down with the velocity less than 0.2m/s, however, rotating or tool joint passing through are forbidden.
- While drilling through pay zone and thereafter, BOP function (close-open) test for the BOPs shall be performed regularly.
- Kill manifolds shall not be used for drilling fluid fill-in.
- The maximum allowed shut-in casing pressure shall be remarkably indicated on choke manifolds.
- All gate valves of well control manifolds shall be tagged with serial numbers, and be indicated with on/Off.
- In case shear rams were used, after shearing the drill pipe with shear rams, retrieve the upper sheared drill pipe, lower a drill pipe joint with TIW valve installed into the hole and close pipe rams on it to prevent the possibility of shear ram's failure.

12.1.2 Management of Well Control Equipment

- Well control equipment used by the drilling crew shall be managed and operated by designated personnel with clear post responsibilities.
- Well site shall have specified component storeroom and rubber element storeroom with air conditioners respectively. The temperature of storerooms shall meet storage requirements of components and rubber elements.
- At least 1 set is available as back-up for every type ram assembly in the well site. Ram preventer side door seals and ram rubber, every type gasket and bolt etc. should be enough for backup.

12.2 Recording Reference

All BOP tests, drills, function tests, any malfunctions, repair or maintenance to the mud System and well control equipment shall be recorded in the IADC daily reports and shall be signed by both the Drilling Contractor's Tool Pusher and Drilling Supervisor on the well site. They shall also be recorded in the Daily Drilling Report.

12.3 Flow Check Procedure

To avoid a kick situation, it is recommended that a flow check be conducted at specific times during the drilling operation.
While drilling (any warning sign).
Prior to tripping the drill string.
While tripping (any warning sign).
While out of hole (continuously).
Trip out:
After pulling 5% of the drill pipe
At the midpoint depth of the wellbore
Prior to pulling the first stand of drill collars
After all the drill string is out of hole.
Trip in:
Upon reaching surface casing shoe
At midpoint of the wellbore

12.4 Primary Control Reference

It is the responsibility of the Drilling Contractor to keep the hole full at all times by using all the available equipment. The Drilling Supervisor shall monitor that the correct preventive practices are being implemented and/or conducted at all times.
Extreme care shall be taken to monitor mud volume, drilling breaks and gas cut mud.
The MACP (Maximum allowable casing pressure) shall be confirmed by the Drilling Supervisor, either after setting each new casing string or, whenever the density of the drilling mud changes.
The MACP shall be clearly written on a kick control sheet that will be posted near the choke control panel.
The predetermined distance at which the Tool Joint is to be pulled above the rotary should be posted on rig floor near the BOP control panel.
The Driller is responsible for carefully measuring and recording the slow pump pressure.

Awareness of these values is an important element in killing operations, in order to avoid formation breakdown.

Take a measure and record the slow pump pressure at three different flow rates with all pumps at the following times:

Once per tour.

Whenever changes occur in the mud density/rheology.

Whenever changes occur in the dimension and characteristics of the string, BHA, jet size, jet plugged or jet lost, etc.

12.5 Secondary Control Reference

Any time a drilling break is noticed, drilling shall stop and a static monitoring of the well shall be carried out. Space-out for all Pipe Rams and Annular Preventer must be known to the Driller at all times to keep tool joints clear of annular preventer and rams.

12.6 Well control during drilling and completion operation

The drilling design shall be strictly implemented by the drilling contractor and when it is discovered there is incompliance of downhole situation with design and the design will be amended, the procedure for amendment of design shall be followed. If emergency is encountered, and safe operation can be affected, after the discussion between drilling rig manager and the drilling supervisor, practical technical measures can be taken first and then report to the top management. The Company shall approve and notice in time as per the drilling program amendment procedure.

The curve of well equivalent mud weight as per predicted formation pressure, curve of mud weight in drilling program and curve of actual mud weight must be drawn during drilling operation. For exploratory well and key well, the curve of dc index while drilling formation pressure equivalent mud weight for sandstone, mudstone and shale shall be drawn by the comprehensive mud logging crew.

When entering oil and gas formation or drilling master reservoir, high diameter nozzle shall be selected for the drilling bit to treat well killing and mud loss.

SCR (Slow Circulation Rate) shall be conducted at the start of drilling activity, pump strokes, pumping rate and circulating pressure should be recorded. While drilling in pay zone, SCR is required to be conducted with all pumps separately as follows, pump strokes, pumping rate and circulating pressure shall be recorded in Daily Drilling Report and Kill Sheet.

Once for every 12 hours in general;

After bit change or BHA change; 3) Mud weight change exceeds $^{3.75}$pcf.

Oil and gas invasion and overflow shall be checked with wiper trip and shall be conducted under the following situation:

Each time before POOH after drilling oil and gas formations.

Severe invasion of oil and gas occurs during drilling, but overflow does not occur before POOH.

POOH after killing well.

POOH after lowering mud weight in wellbore.

POOH after curing mud loss during drilling oil and gas formation.

POOH after drilling oil and gas formation and a prolonged period of time is needed for other operations (E logging, running casing, running tubing and DST etc.).

Regarding the shut-in methods selection

Soft shut-in method is recommended for gas well;
Hard shut-in method is recommended for oil well;
Hard shut-in method is recommended for the well of which the well has strong formation strength in EMW as per FIT test;
Soft shut-in method is recommended for the well of which the well has weak formation strength in EMW as per FIT test.

Main technical measures for prevention of kicking and blowout during tripping.

Before POOH, circulation duration of mud in wellbore shall not be less than one and a half cycles. The observation time after wiper trip shall be no less than one and a half cycles. The density differential of inlet and outlet shall not exceed 1.248 pcf.
Flow check shall be conducted regularly via trip tank as follows
Prior to POOH (keep string rotation at 10 rpm for preventing stuck pipe)
POOH to last casing shoe
Before POOH BHA through BOP stack
Note: Tripping should be shut-off immediately and flow check shall be performed later on when any suspected well kick is observed.
During POOH, fill up hole every 3-5 stands of DP and every stand of DC. If the nozzle is plugged, during POOH, mud shall be filled full for each joint or continuous filling shall be conducted and trip sheet shall be recorded carefully.
When the bit is in the oil and gas formation and 300m above top of oil and gas formation, POOH speed shall be controlled to maintain the excellent well building property and rheological property of mud to avoid severe swabbing in the wellbore.
During RIH, the running speed shall be controlled to avoid the loss of circulation due to down hole pressure surge.
During tripping, check and repair of the equipment shall be arranged at the time when running to the casing shoe. if equipment have to be checked and repaired during POOH, due blowout prevention measures shall be taken during and after the check and repair, drill string shall be run to bottom and circulate for one and a half cycles. After everything is normal, drilling shall be resumed. Equipment check and repair shall not be conducted for empty well condition.

Treatment measures for invasion of gas during normal drilling:

When invasion of gas is discovered, it shall be discharged with circulation in time. The gas invaded mud shall not be pumped in the hole before eliminating gas.
Increasing mud weight shall be conducted when stop drilling and after invaded gas eliminated from mud.
Mud weight incremental of each circulation cycle shall be controlled within 3.121 pcf

"In-post" policy

"In-post" will be initiated after the surface casing is cemented.
The "in-post" personnel must be trained and verified qualified by the technical staff of drilling rig.
During drilling, specific personnel shall be arranged to be "in-post", the mud logging personnel shall monitor the level of the circulating tank etc. as per the requirements. During tripping, other auxiliary

operation or drilling suspension, specific personnel for "in-post" shall be arranged by the drilling and mud logging personnel in the meantime.

"In-post" requirements:

For drilling operation, variation of the outlet flow, drilling time, lithology, bubble, smell, showings, pit volume, density and mud viscosity, gas measured values and content of chloride etc. shall be carefully monitored. The level of the circulating tank shall be observed and recorded each 15 minutes and if special situation occurs, the measurement and record shall be intensified. If anomaly occurs, it shall be reported to the driller in time.

For tripping operation, trip sheet shall be filled accordingly. If anomaly occurs, it shall be reported to the driller.

For other operations, it shall be observed whether there is overflow of mud at the wellhead or mud level from wellhead. If anomaly occurs, it shall be reported to the driller.

Treatment of loss of circulation

Before cementing non-target zone or pay zone, if loss of circulation occurs during drilling, POOH shall be carried to run open ended drill pipe for cement plug job if condition allows. During RIH, mud shall be filled from annulus to balance formation pressure to avoid blowout.

Treatment of loss of circulation during drilling after production casing cemented

Mud level not at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and observe with moving the drill string and prepare well control related work. Each 10 minutes, mud with density no lower than the drilling weight shall be reversely filled up from the annulus. Each time, the filling up shall be 1~2m^3, or continuous reserve injecting mud with small speed to maintain that the column pressure of mud in wellbore is higher than the formation pressure. POOH to casing shoe or safety interval for observation. Additional treatment shall be made as per the situation of loss of circulation.

Mud level at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and move the drill string to observe with well control well prepared. When the well is static, pull the bit upward to the casing shoe or safe interval for observation. If no anomaly is observed, circulate with slow pump rate. The measures such as plugging back job, POOH or condition mud etc. shall be carried out as per the situation of loss of circulation. If loss of circulation occurs to pay zone, mud weight shall be acquired by static pressure. After pulling out to casing shoe and observing for one tripping cycle, when no more loss of circulation occurs after running to bottom, drilling operation can be resumed.

Well control for E logging, cementing and DST

Before E logging, down hole condition shall be stable and static. If it takes long time for electrical logging, one more condition trip shall be conducted between logging. Shooting nipple is required for logging. Logging service shall be furnished with tools for cutting the cables.

Continuously fill the well during performing wireline logging, and keep closely monitoring for the well via trip tank.

Before running casing, ensure the well is in static condition without any flowing. Pipe ram shall be changed to casing ram with proper size. Casing circulating head or crossover sub between casing and in used DP shall be kept at rig floor at all times during casing running.

The hydrostatic pressure shall be kept balance with formation pressure during whole cementing job to avoid blowout due to loss of circulation, loss of balance for pressure and cement weight loss during WOC.

12.7 HEALTH

This well is committed to develop for oil in a responsible manner that promotes employees' and local inhabitants' health.

A trained Paramedic and proper MEDIVAC equipment and ambulance shall be stationed on the drilling rig during the drilling campaign.

An emergency medical evacuation plan shall be coordinated, and shall keep in touch with emergency response centres manned by qualified doctor 24-hours per day in Ahwaz city.

Medical emergency report and investigation shall be performed as soon as possible in accordance with Medical Emergency Procedures.

12.8 SAFETY

Due to high concentration of H2S in Sarvak, Kazhdumi and Gadvan reservoirs, Make detail H2S operation procedures, in particular, and carry it out strictly.

Lifesaving equipment like SCBA, Escape Mask, H2S detectors (Multi portable and personal detector) and implement the H2S Services system during drilling activity should be provided.

All lifting equipment and cranes must be certified by approved 3rd party (approved by Institute of Standards & Industrial Research of Iran)

All electrical equipment should be designed for such locations, and listed by nationally recognized testing laboratory. All wiring components and electrical equipment should be maintained in accordance with the manufacturers' recommendation.

PSSR (Pre-Startup Safety Review) before spud-in and penetrating into oil and gas formations are mandatory by contractor under the supervision of Company and approved by Company.

12.9 ENVIRONMENT

Requirement on location

The access road and well location construction shall be accorded with construction program of well location. It minimizes any impacts on the local residents, prevent all soil contamination, prevent all ground water contamination, and prevent all surface water contamination. Cutting and waste water pit shall be lining and drainage system shall be good and to avoid farm and season canal contamination.

Management regulation for transportation truck, drilling operation machine and tools shall be carried out strictly to avoid damage farm and trees, etc.

Well site shall be flat and clean. All the equipment, materials, drilling tools shall be put in order.

Control the rig noise to decrease the influence to surroundings while drilling at the inhabitation area.

Contractor shall perform on-site segregation of wastes prior to transfer to the transfer area and/or delivery to the transportation Contractor in accordance with the following waste hazard classification.

Non-Hazardous Wastes (Re-usable, Recyclable and Recoverable Wastes)

Ferrous and non-ferrous metals, plastics, and wood wastes, PETs and potentially paper and glass.

Materials shall be separately segregated and provision may be made for size reduction and containerization equipment.

The Contractor/Generators are encouraged to seek for recycle/re-use alternatives based on the Local available facilities, for final disposal of these materials.

It should be noted that landfill disposal of metal waste materials and wood wastes shall be prohibited. For wood wastes re-use of and/or salvage of commercial timber, or deliver/release to.

Local community are recommended.

Hazardous Wastes

Hazardous Waste, such as clinic waste, oil waste, waste produced by painting, insulation works shall be duly segregated.

Facilities of environment protection supplied to well site

Mud purifying circulation system.

Cooling water spraying and circulating system for drilling pump.

Special recovery tank for deposed oil.

Reserve house for valuable mud chemicals.

Automatic mud weighting supply system.

BOP stack is equipped according to drilling design.

Define a separation area for gathering all waste according to waste management procedure

Requirement of environment protection on location

Sewage pit Area of the sewage pit: 1500 m2.

Prior to spud, mud pit and sewage pit must comply with the standard. Install cooling water spraying and circulating system for drilling pump and sewage circulating system for cleaning rig floor and equipment.

Install BOP stack and pressure test shall be operated according to the drilling design to avoid blowout.

Install oil supply lines and check whether there is any leakage.

Management of environment protection on location

In drilling operation keep mud and sewage from overflowing. If it happens, clean up timely.

Mud chemicals shall be stored according to the standard and shall not be scattered around well site. If mud chemicals scatter while loading or unloading, clean it up at once. The sack of mud chemicals shall be recovered. Do not dispose it on site.

Enhance water management at well location and do not wash equipment with fresh water. Stop water flowing for long time. Overhead tanks shall be utilized.

Oil management shall be carefully at the well site. No oil dirt on site, no dripping and leaking at oil tank and manifold system area.

After the well is completed, reinforce the sewage pit solidify the disposed mud and have environment procedure fulfilled.

Install fuel supply lines and keep them in good condition.

All the drilling fluid materials shall be recovered and shall not be left at the well site after completion.

The environmental protection personnel from contractor and operator side shall check and accept the well site environment according to the regulation after completion. Only well site environment is qualified, the well can be delivered.

Transition Area for Hazardous Waste shall also be provided by each Contractor in his allocated area, Company shall approve each area before use as well. The transition area of hazardous waste shall be:

Paved, fenced, marked and illuminated

Equipped with appropriate Fire Extinguisher and other emergency response equipment (e.g. Eye-Wash and Emergency Shower, appropriate First Aid kits, neutralizer substances etc.).

In compliance with the requirements of handling, transportation and storage of hazardous waste (based on usage chemical MSDS).

Configured to accommodate a safe physical separation of incompatible waste materials (e.g. acids, and alkalis, organic materials and strong oxidizing agents).

Capable enough to provide storage for the entire range of hazardous wastes generated during the work.

The area shall be provided with Spill Containment and means of preventing run-off to and from the area.

All waste containers to be stored shall be appropriately labelled.

Hazardous Waste shall be transferred in a timely manner, within a maximum period of 48 hours, for any period in excess of this permission shall be granted from Company and/or PEDEC.

13. H_2S Monitoring & Prevention Measure

13.1 Monitoring H2S

Somewhere on the well site is apt to accumulate H2S, such as mud pit, shale shaker, and drilling platform, should be installed one set of. Make sure H2S monitor and alarm system can be used. Alarm threshold setting for H2S warning indicator: the first warning level is 10mg/m3, unable warning sound, just show present value of H2S content to operators. The second warning level is 20mg/m3, when the H2S content in air exceed 20mg/m3, H2S warning indicator can give an alarm automatically, and warning sound should be heard by everyone work in well site. The audible alarm should be installed on Kelly board.

Equip enough gas masks on drilling platform, shaker screen (total amount should large than 24), and equip 12 sets of 30min positive pressure Air Aspirators 2 sets of portable H2S monitor and appropriative H2S warning indicator. Arrange a person specially assigned for taking charge all the protective apparatus. This man must do periodical check to keep these equipment in a good condition.

H2S monitor should be done weekly check and compulsory examination.

After drilling into the pay zone, strengthen the detection of H2S content in mud.

13.2 Measures for H2S prevention

Windsocks shall be set on the drill floor and in the monsoon direction or/and prevailing wind direction of Azadegan oil field, off well site. In emergency, employees must follow the evacuation procedure.

Explosion-proof blowers shall be installed around the rig, shale shaker, and drilling fluid tanks to blow away toxic and flammable gas at the well site.

The rig crew in the H2S area shall be furnished with H2S monitors and adequate protective apparatus. Everybody shall be able to use, maintain, and check the apparatus.

The technicians of rig crew are responsible for H2S proof training and the tool pusher is in charge of the supervision and examination.

The pH value of drilling fluid in the H2S area shall be kept over 9.5. Pay attention to measurement of H2S concentration in drilling fluid. The H2S concentration shall be kept at less than 50mg/cm3 by using sulphur removal agents and degassers.

Personnel safety and protection shall be carried out according to SY/T5087 during operations in polluted areas where H2S content exceeds the critical concentration.

When the rig crew makes decision to bleed and fire because well control cannot be performed due to well site conditions, operations shall be performed according to procedures in SY/T5087.

Once blowout occurs, the incident shall be reported to the higher level department as soon as possible and firefighting trucks, ambulances, medical care personnel and technicians shall be on duty at the well site.

After the blowout is controlled, H2S concentration should be measured in the places where H2S is possible to gather.

The personnel are not allowed to access until the H2S concentration is below the critical concentration.

14. Handover Data

The handover document should contain the following well construction information:

- Pre-spud Inspection Report
- Daily Drilling Report
- Daily Mud Report
- Daily Mud Logging Report
- Completion Tally
- Completion running Report
- TMS data (Completion)
- Dry Test report
- Test Charts, X-Mass tree Installation & test Report.

15. Requirements

Item	Description	QTY.	Unit	Supplier
1	Bit 8 1/2" Mill Tooth	1	ea.	PPL
2	Bit 5 7/8" Mill Tooth	1	ea.	PPL
3	4 1/2" * 3 1/2" Completion String 10K	1	set	PPL
4	X-mass tree 11" * 4-1/16" 10K	1	set	PPL
5	THS 13-5/8'' 5k* 11'' 10k	1	ea.	OEOC
6	9-5/8" & 7" Scraper	1+1	ea.	OEOC
7	5" DP,HWDP,8-1/2" DC,	as required		OEOC
8	Well Head Services	as required		Service Co.
9	Surface Well Test Services	as required		Service Co.
10	Completion Services	as required		Service Co.
11	Slick line Services	as required		Service Co.

12	Casing Running Services	as required		Service Co.
13	Wireline Logging & Perforation Services	as required		Service Co.
14	Dry Test,DST,RTTS	as required		Service Co.
15	Cementing Services	as required		Service Co.
16	Stimulation (Coiled Tubing, Acidizing and N2) Services	as required		Service Co.
17	H2S Services	as required		Service Co.
18	Mud Logging Services	as required		Service Co.
19	Rig Services	as required		Service Co.
20	Drilling & Completion Fluid Services	as required		Service Co.

Note: Quantity & Spec to be conformed with contract's SOW.
Note: In case of any shortage or discrepancy, the contract's SOW will be reference.

Table 20: List of Requirements

16. Revision Corrections

Drilling Engineer & Geology							
1	جدول Logging Program گزارش مطابق با شرح کار باشد **(جدول زیر)**. نمودار CCl/GR یا MTT بعد از مشبک کاری اضافه شود. متن زیر به صورت Note در زیر جدول Logging و سکانس عملیات اضافه شود: In case of not confirming the perforated intervals by the CCl-GR log, the Contractor should use MTT logging tool.	25	CM	• Logging Program بر اساس شرح کاری است که پیوست قرارداد اصلی می باشد و تغییر آن منوط به ابلاغ در قالب اضافه کار می باشد.			
2	در بخش هایی از برنامه نام چاه AZNS-10 و در بخش هایی AZN-10 نامگذاری گردیده است. نام صحیح AZN-10 می باشد. اصلاحات لازم انجام گردد.		CM	• اعمال گردید.			